Meta-analysis, a very, very, very small introduction

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EXCELSCIOR

SORTEE
Society for Open, Reliable, and Transparent Ecology and Evolutionary Biology

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- Ornithologist by training









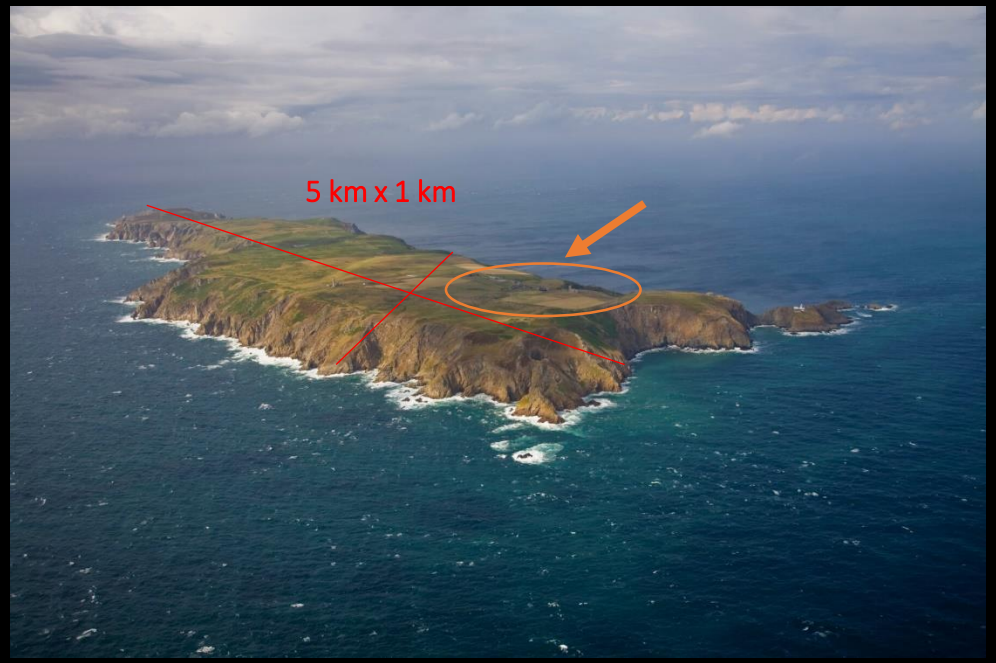






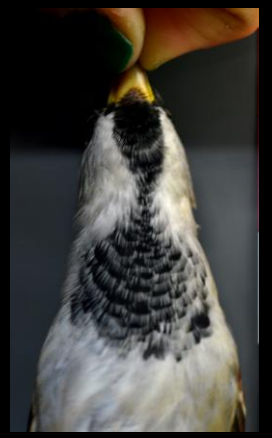
Status Signalling Hypothesis / Badges of status ?

- ✓ 4+ years
- ✓ Analysed 20,000 sparrow fights

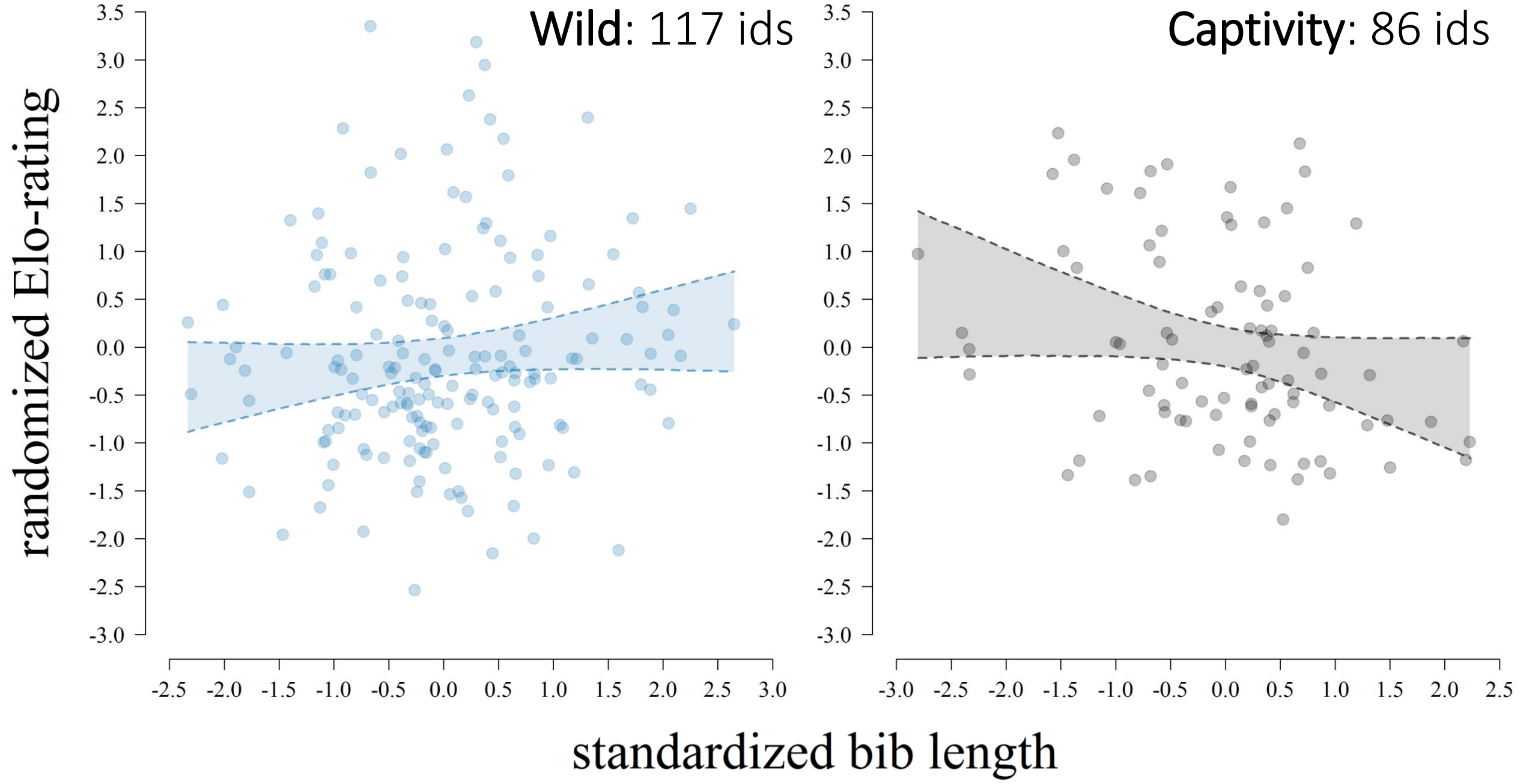











Wild: 117 ids
Captivity: 86 ids
randomized Elo-rating
standardized bib length
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
-0.5
-1.0
-1.5
-2.0
-2.5
-3.0
-2.5 -2.0 -1.5 -1.0 -0.5 0.0 0.5 1.0 1.5 2.0 2.5 3.0
-3.0 -2.5 -2.0 -1.5 -1.0 -0.5 0.0 0.5 1.0 1.5 2.0 2.5

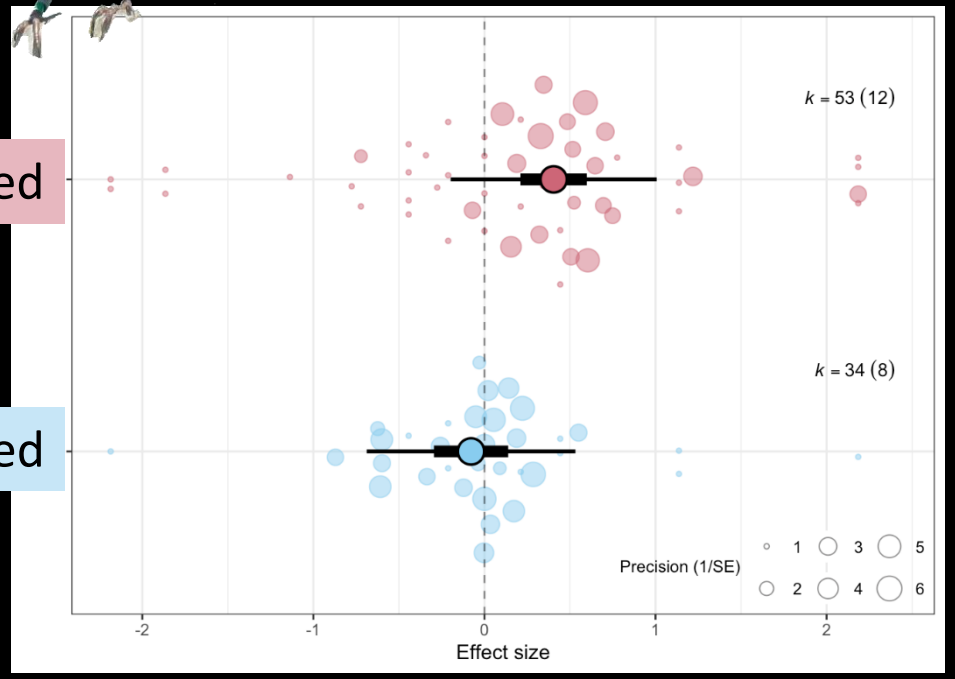
Oh my!

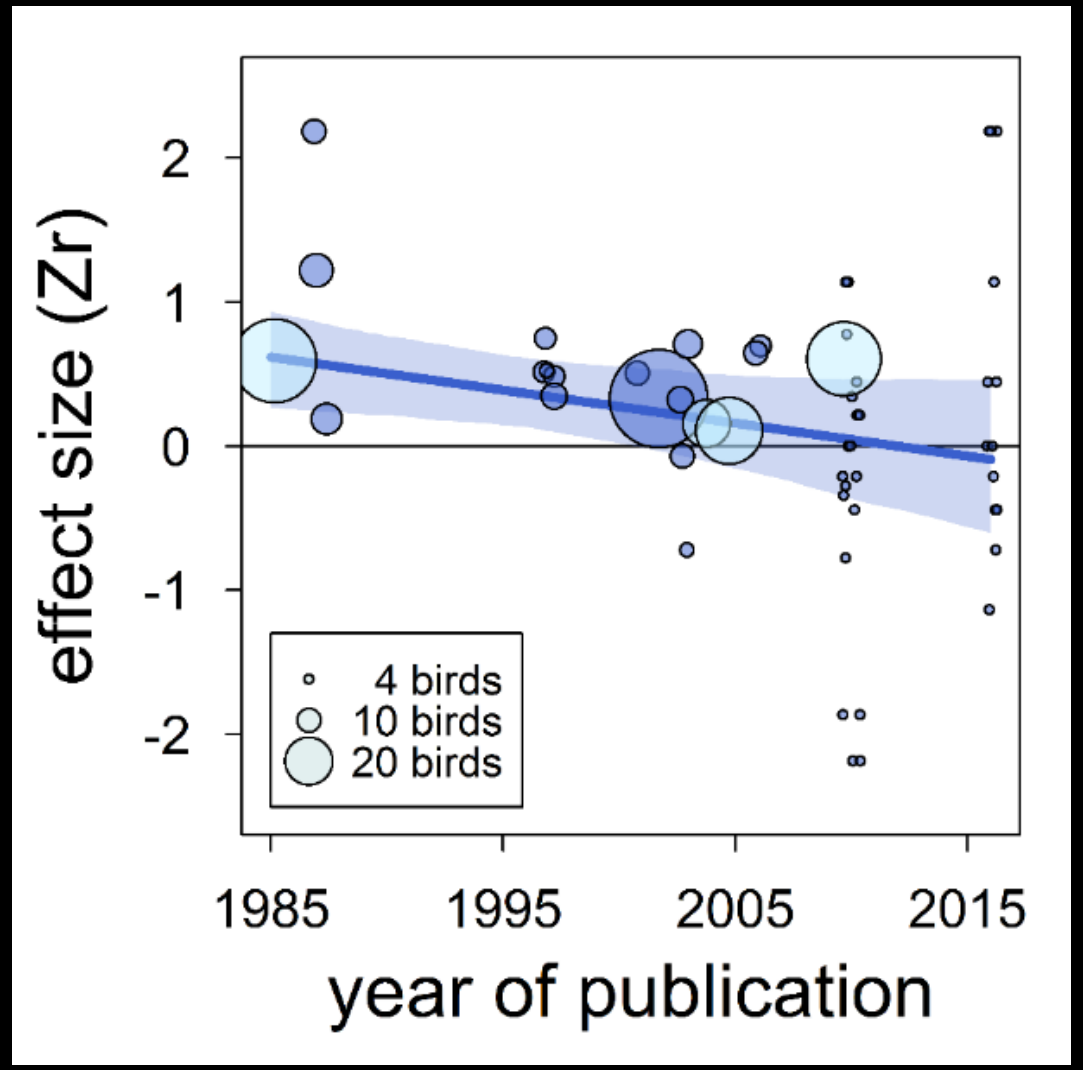


$R^2_{marginal} = 41\%$

Published

Non-published

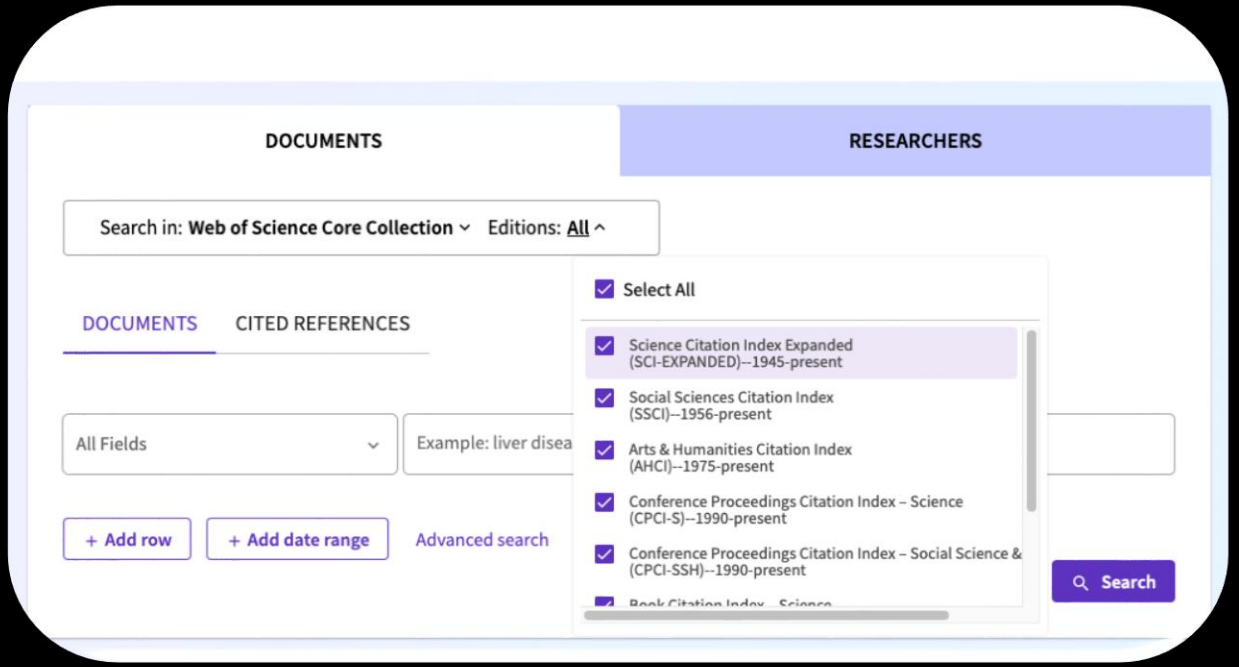




What is a meta-analysis?

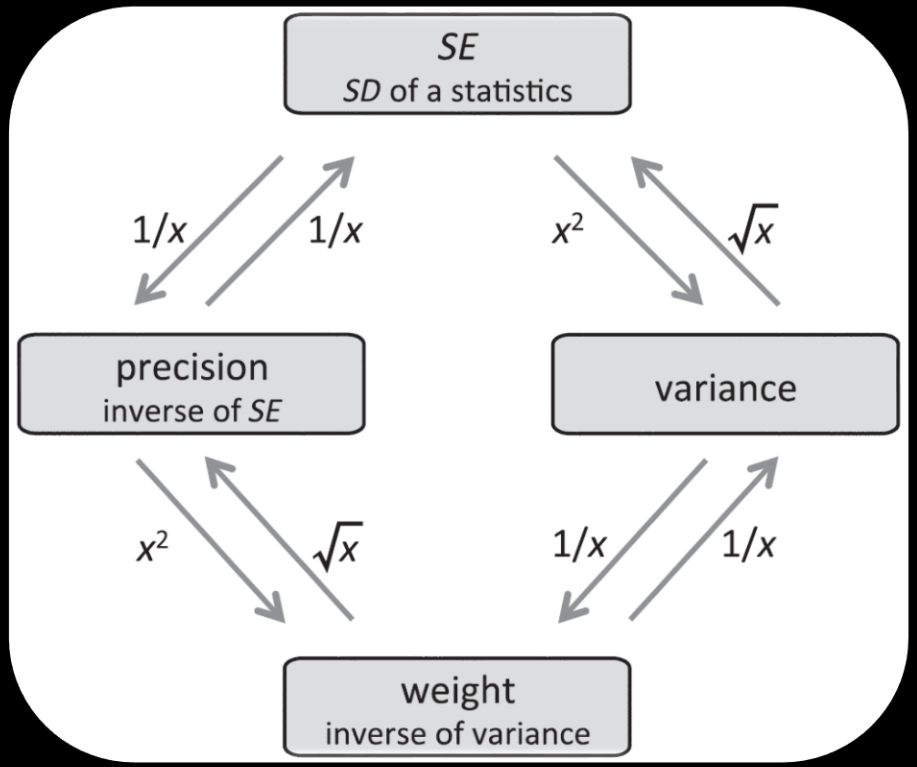
What is a meta-analysis?

- Preceded by a Systematic Literature Review



What is a meta-analysis?

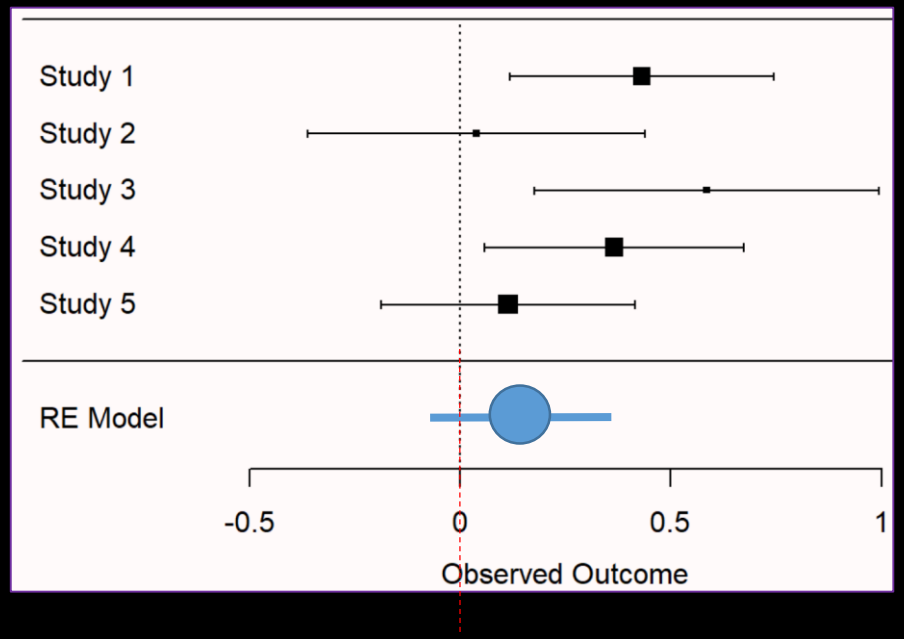
- Preceded by a Systematic Literature Review
- Quantitatively combines evidence (weighted)



Nakagawa *et al*. 2022, MEE

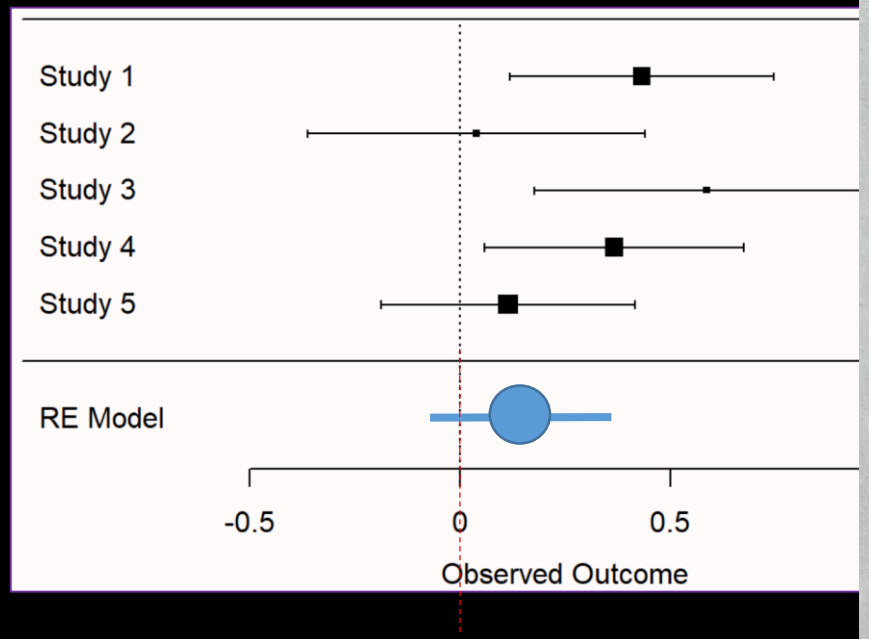
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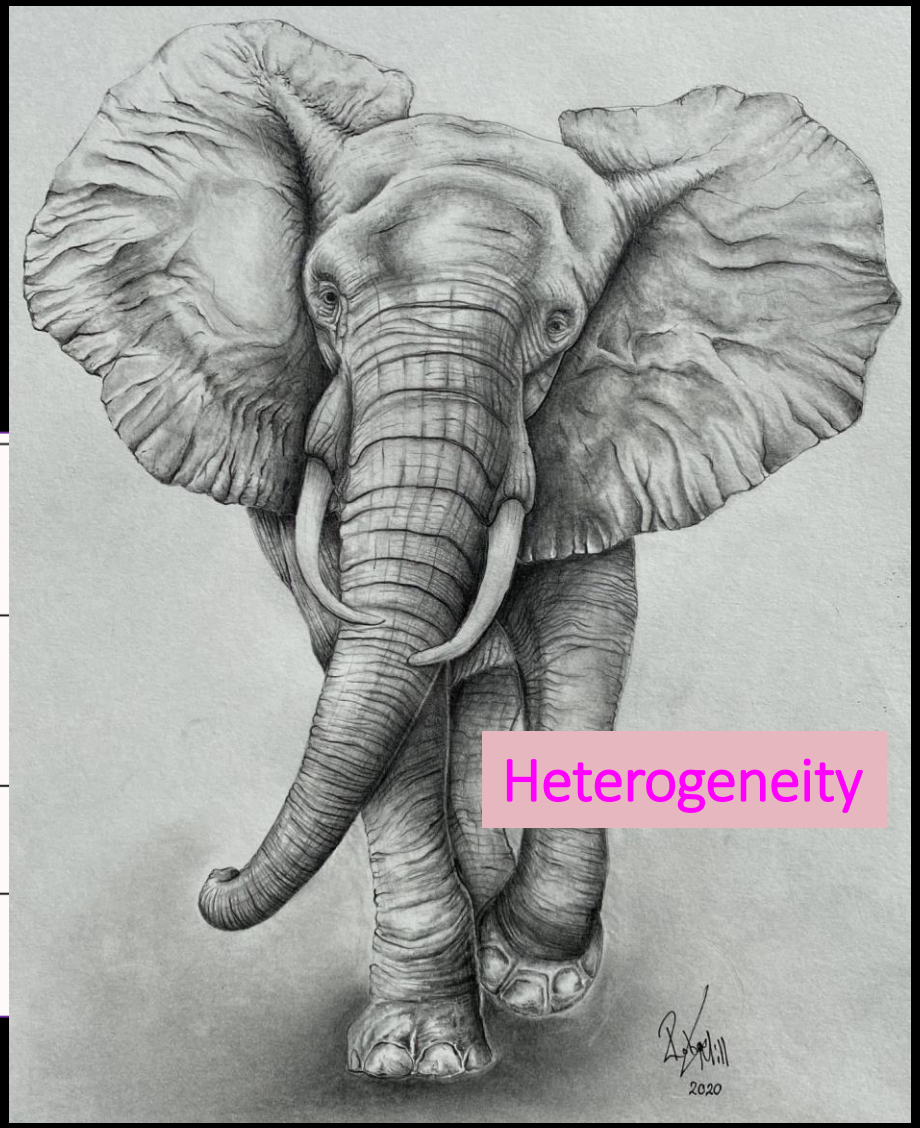
- Test overall effects



What is a meta-analysis?

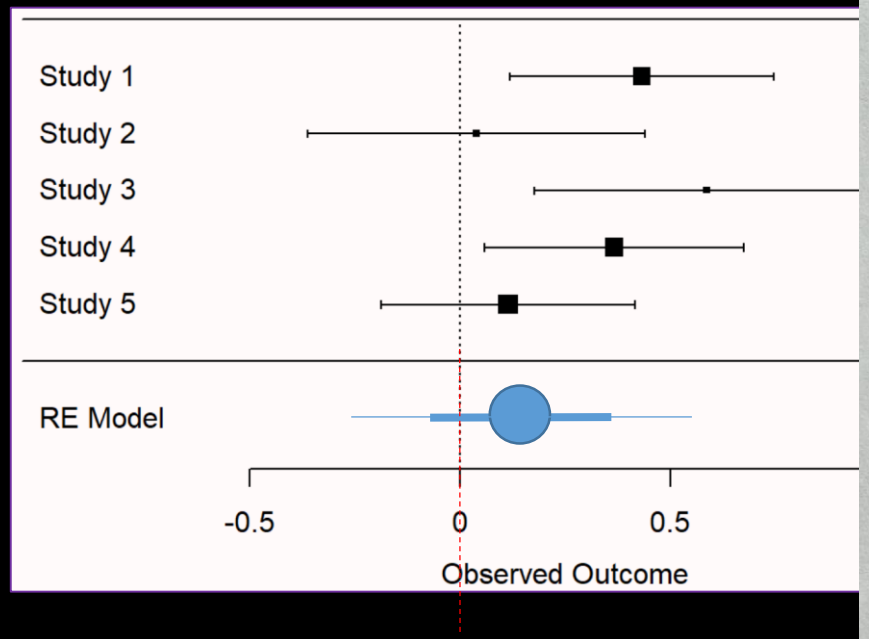
- Goals:
  - Test overall effects
  - and their generality

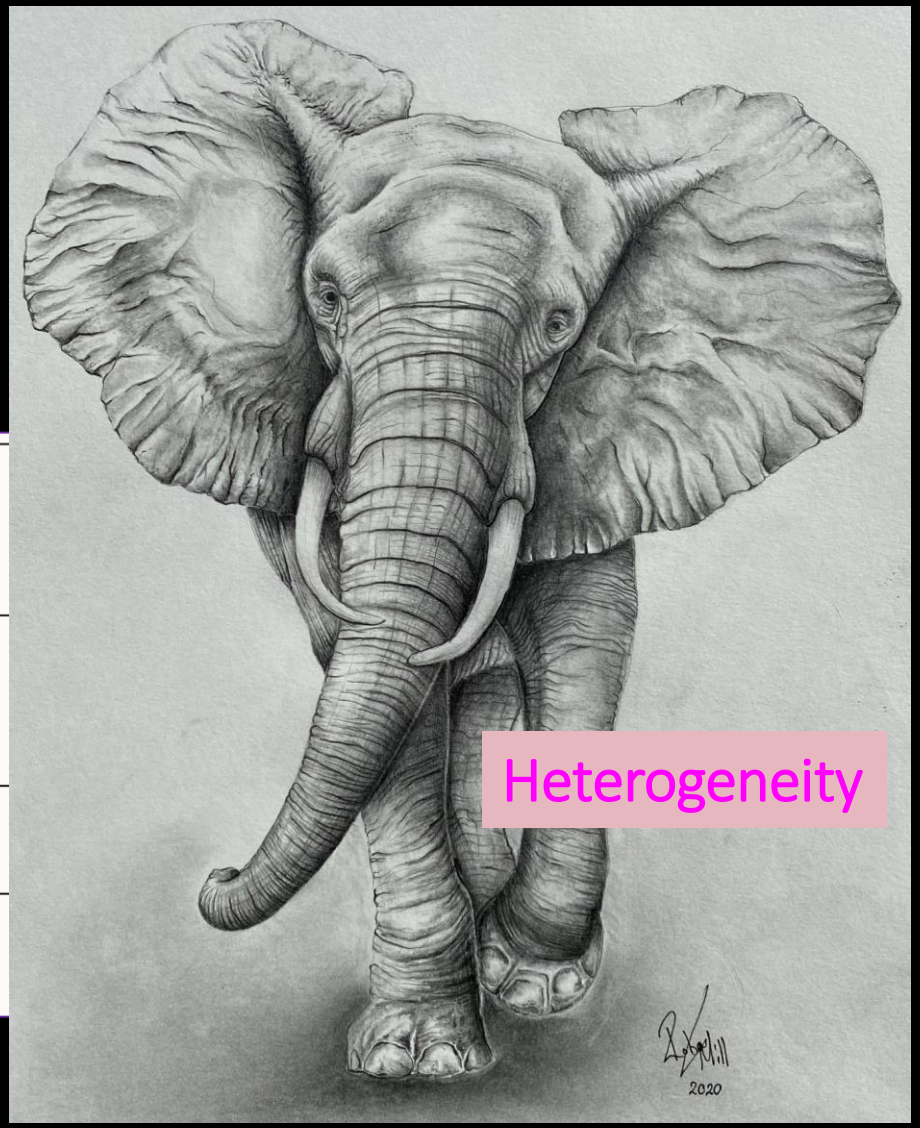




What is a meta-analysis?

- Goals:
  - Test overall effects
  - and their generality

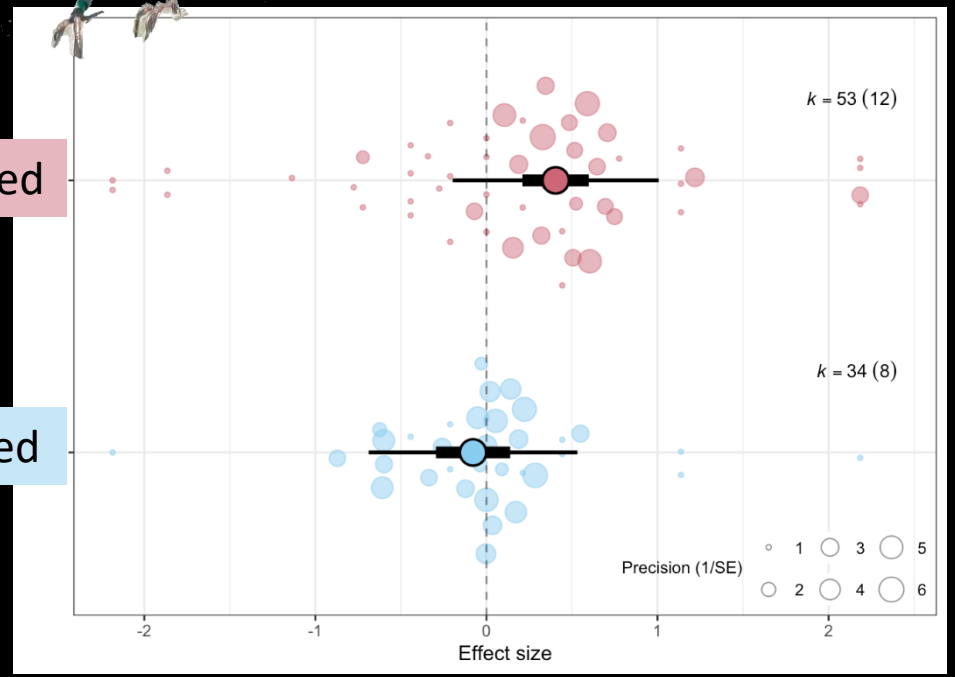




What is a meta-analysis?



$R^2_{marginal}$ = 41%



- Goals:
  - Test overall effects and their generality
  - Understand pattern of effects

→ Meta-regression

Sánchez-Tójar *et al*. 2018, eLife

The Funnel plot

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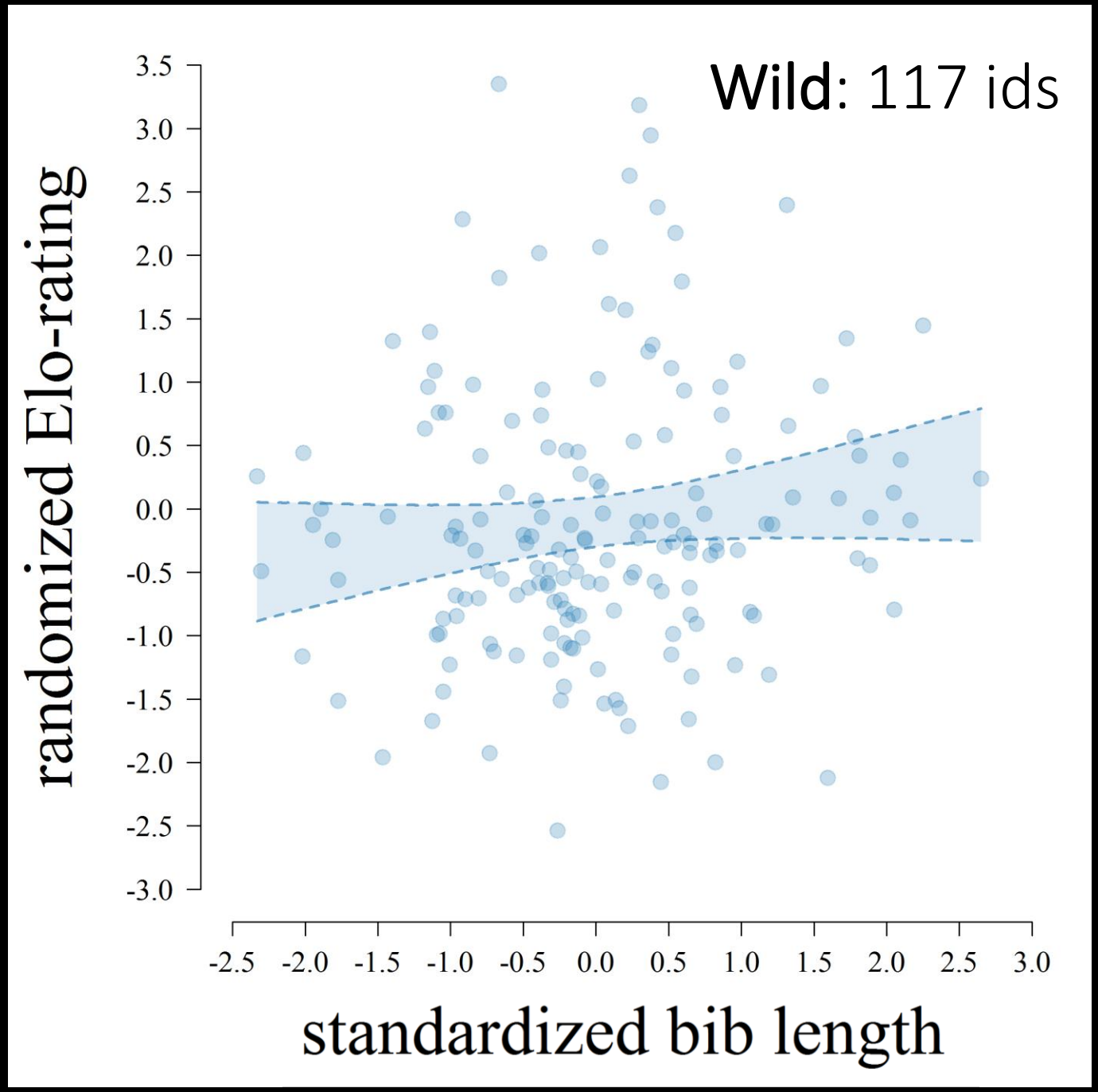
EXCELSciOR

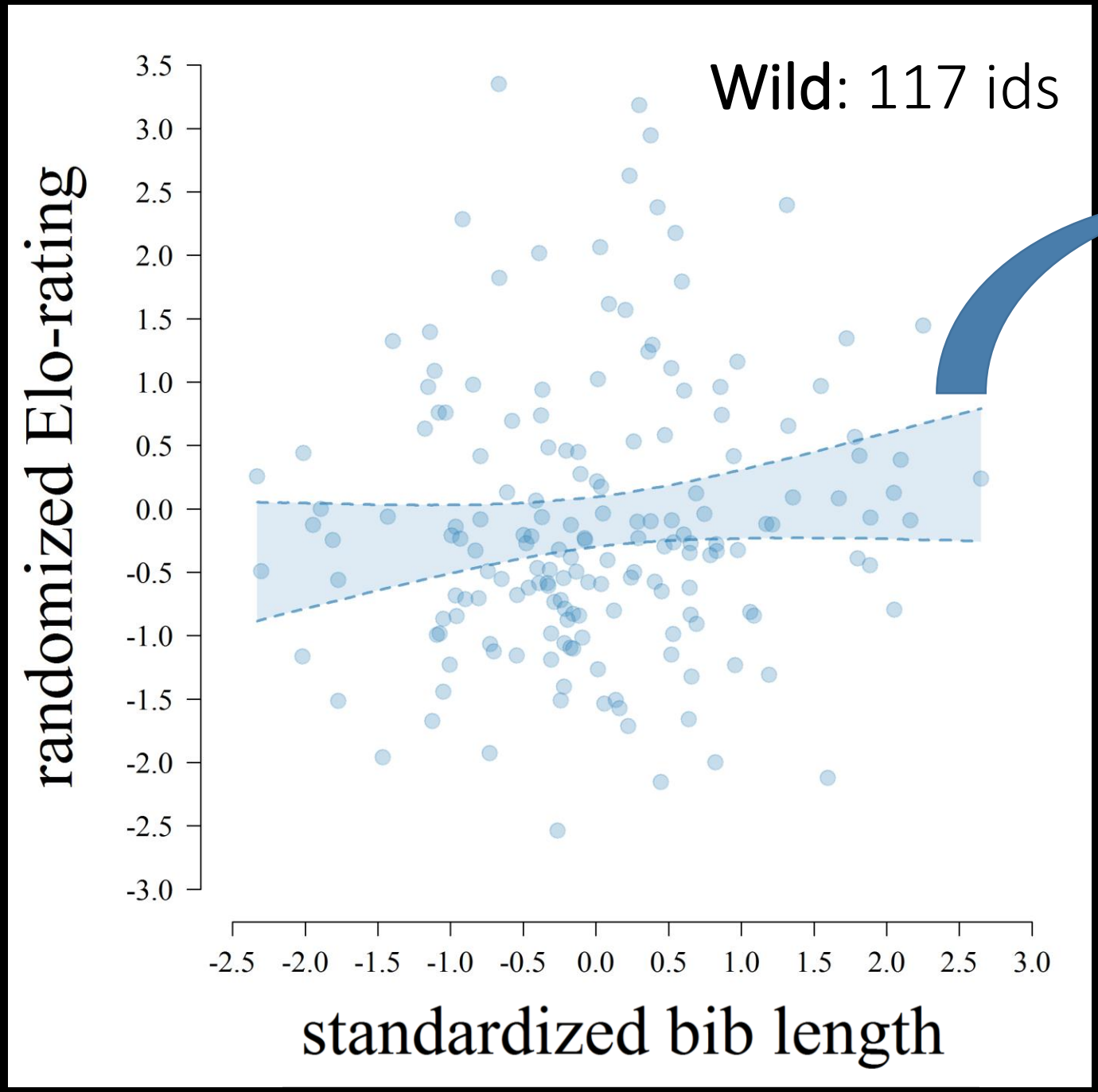
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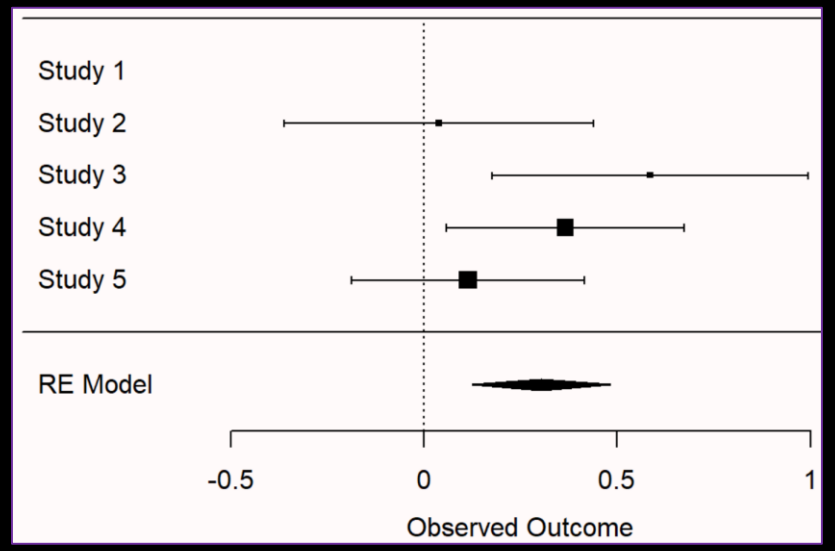
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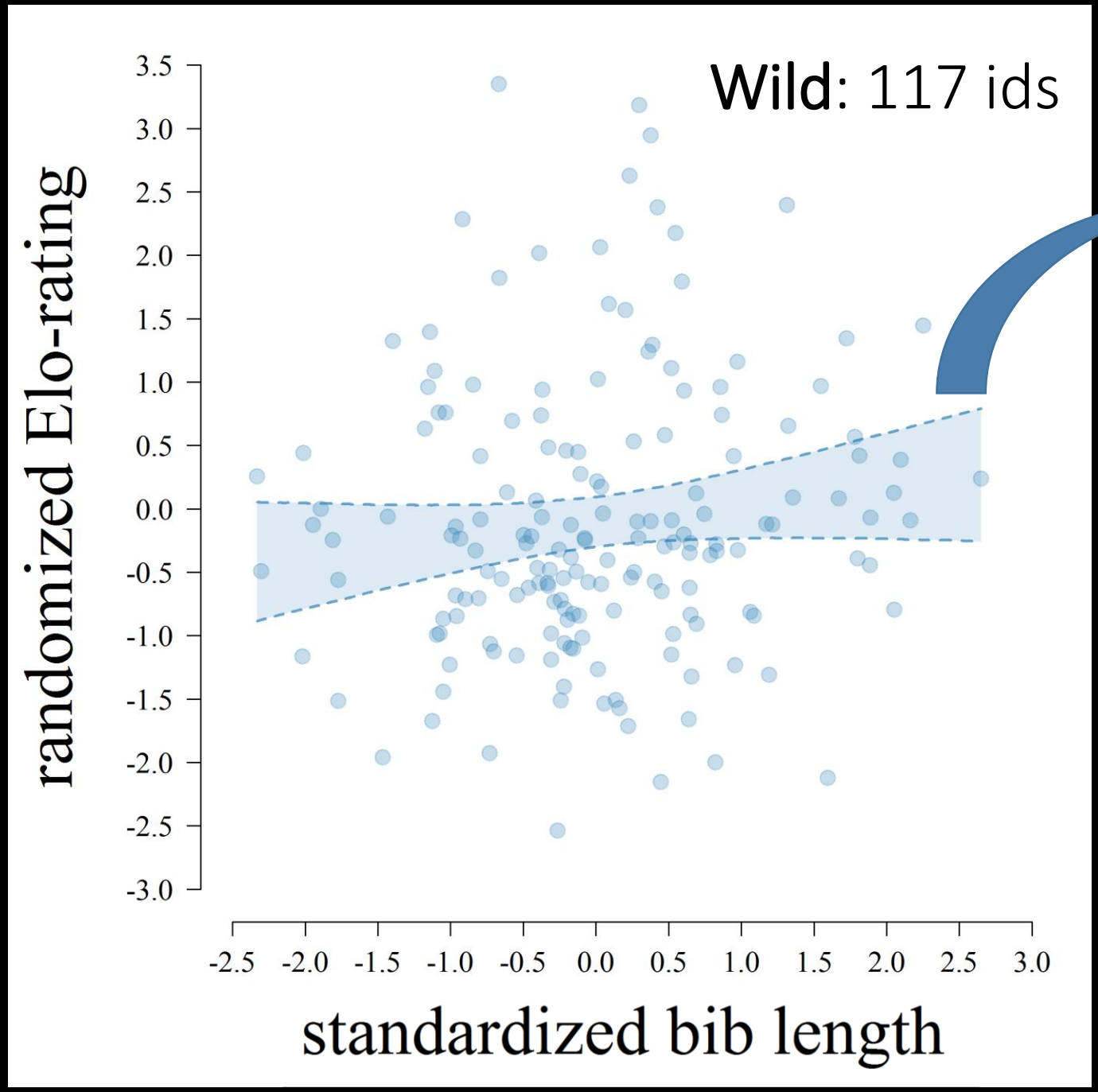
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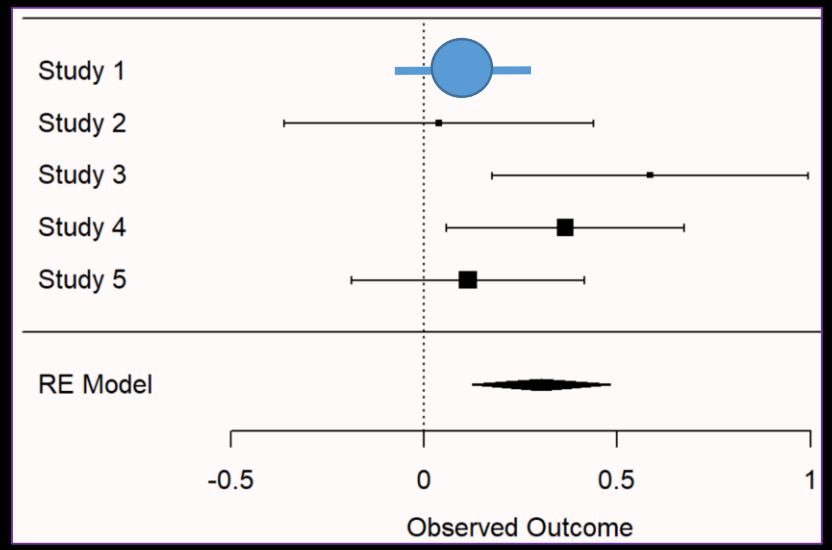

Wild: 117 ids
randomized Elo-rating
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
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-1.0
-1.5
-2.0
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-3.0
-2.5 -2.0 -1.5 -1.0 -0.5 0.0 0.5 1.0 1.5 2.0 2.5 3.0
standardized bib length

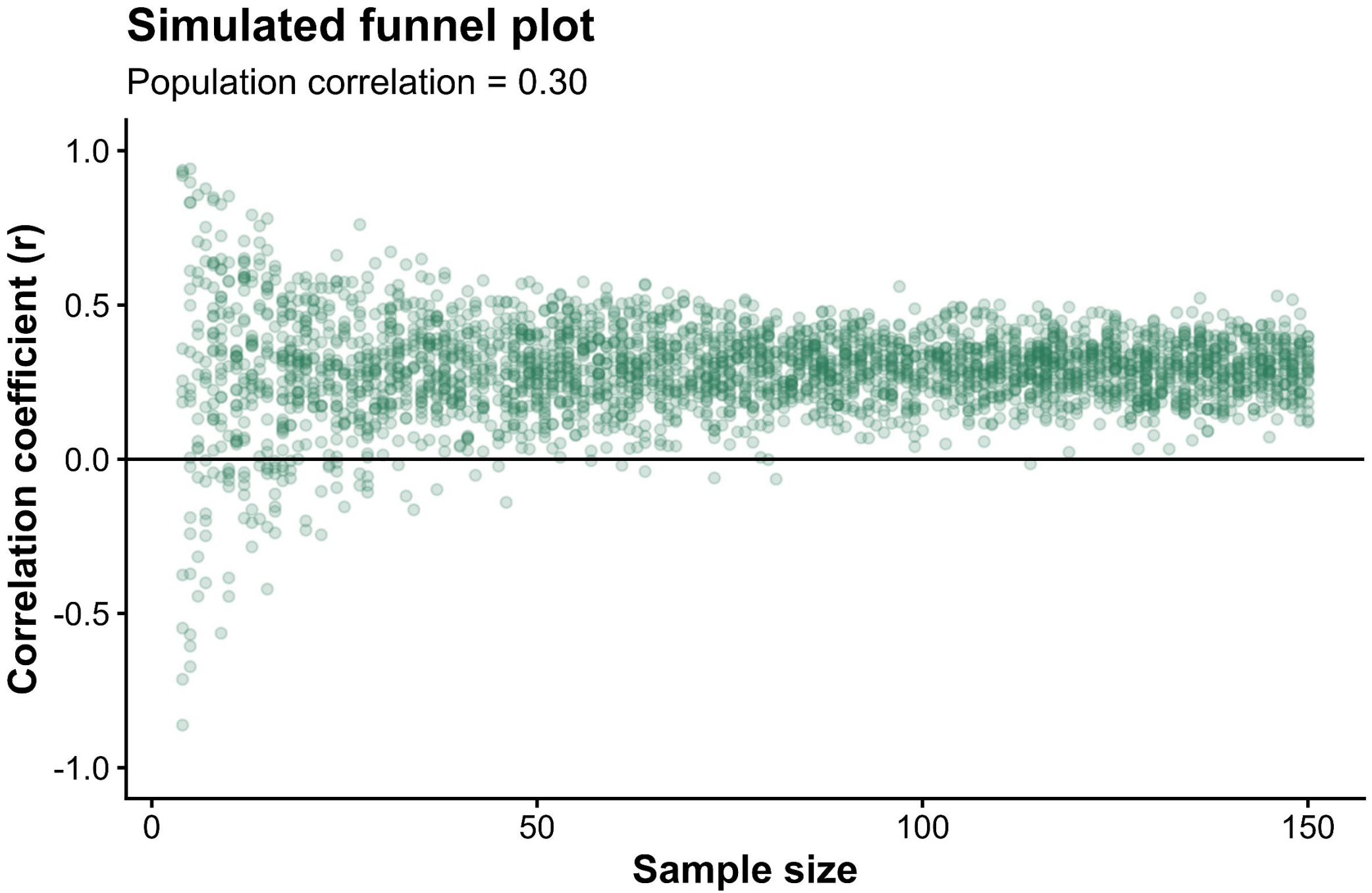

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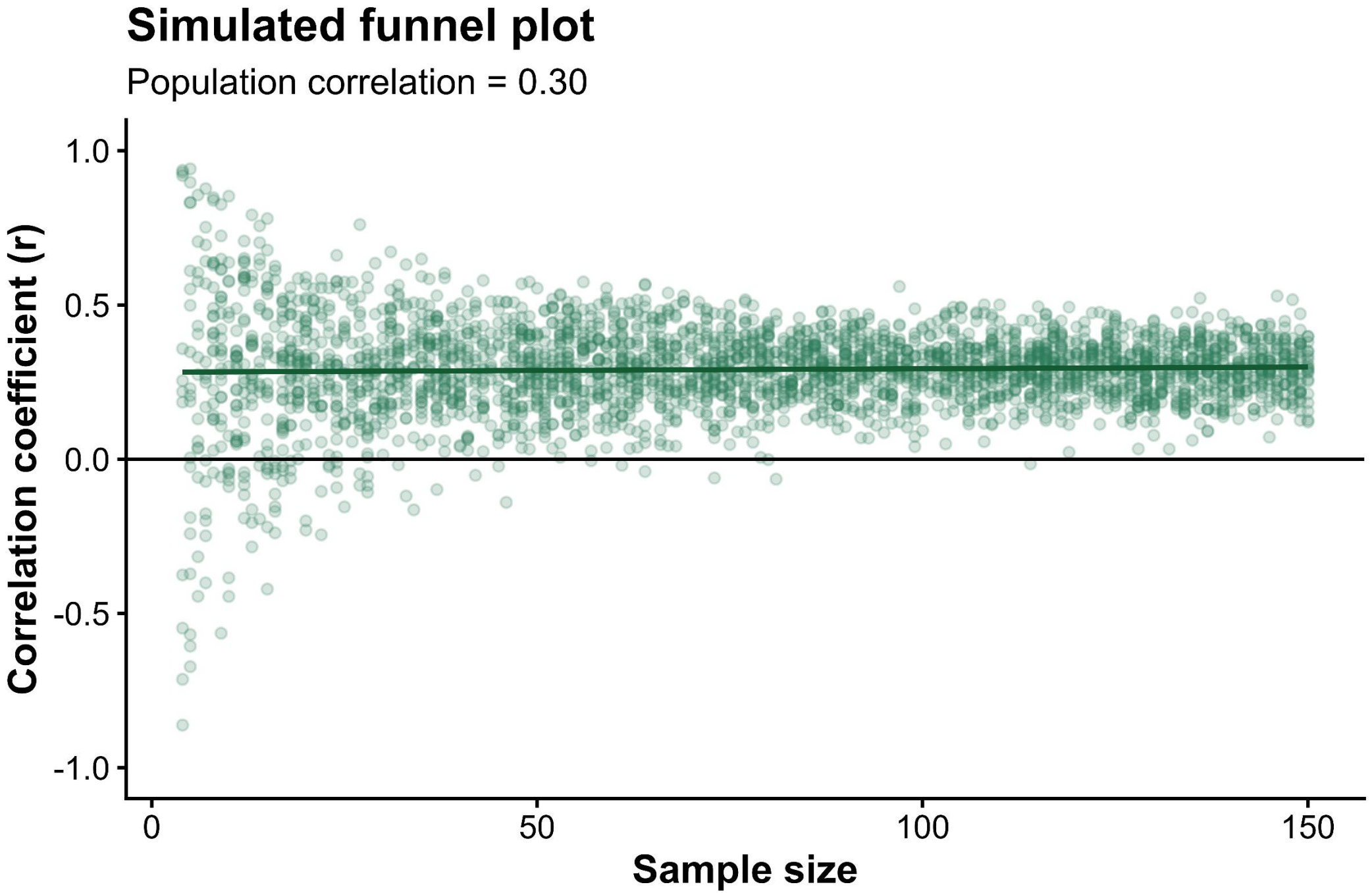

Study 1
Study 2
Study 3
Study 4
Study 5
RE Model
-0.5
0
0.5
1
Observed Outcome

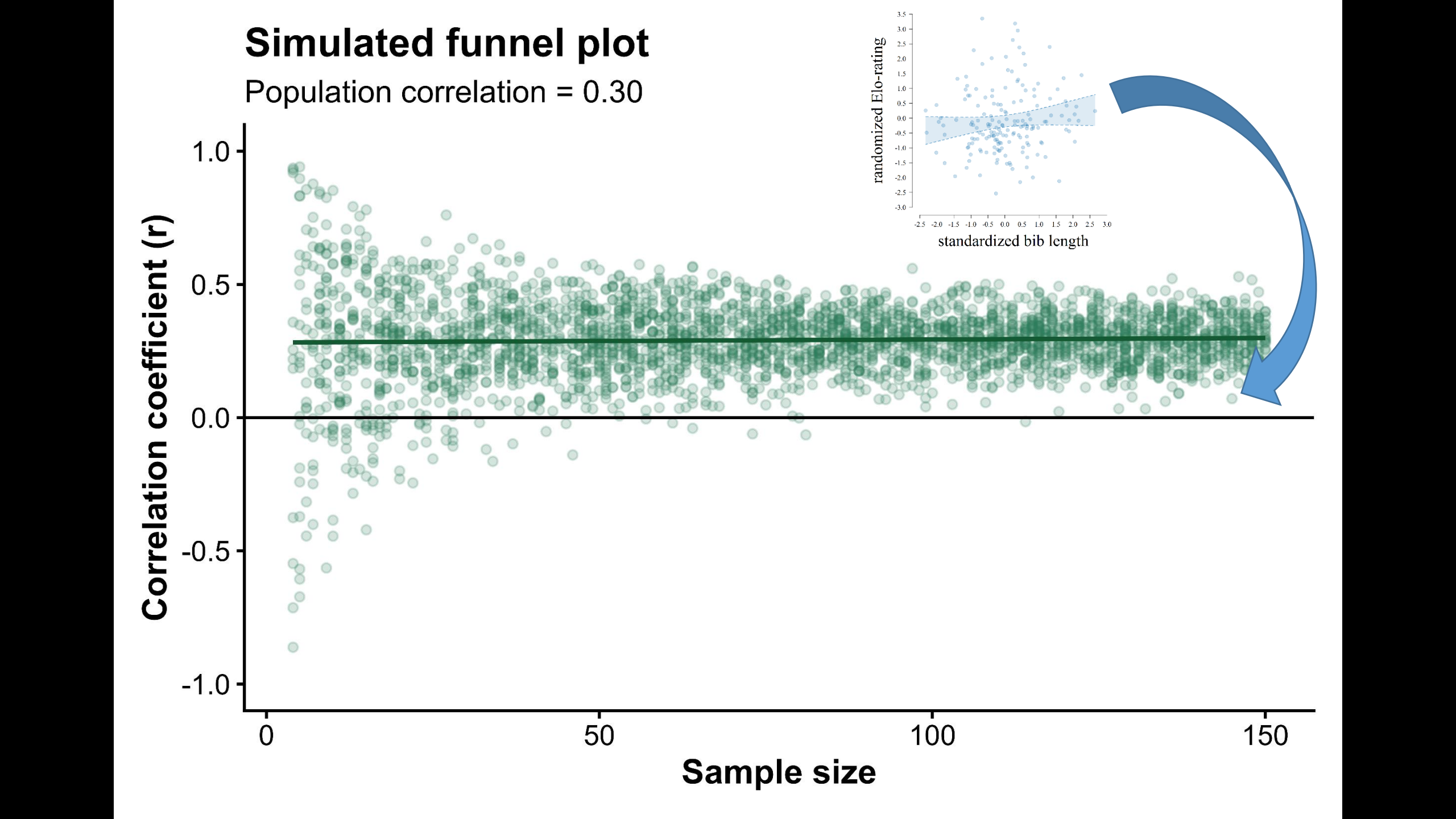


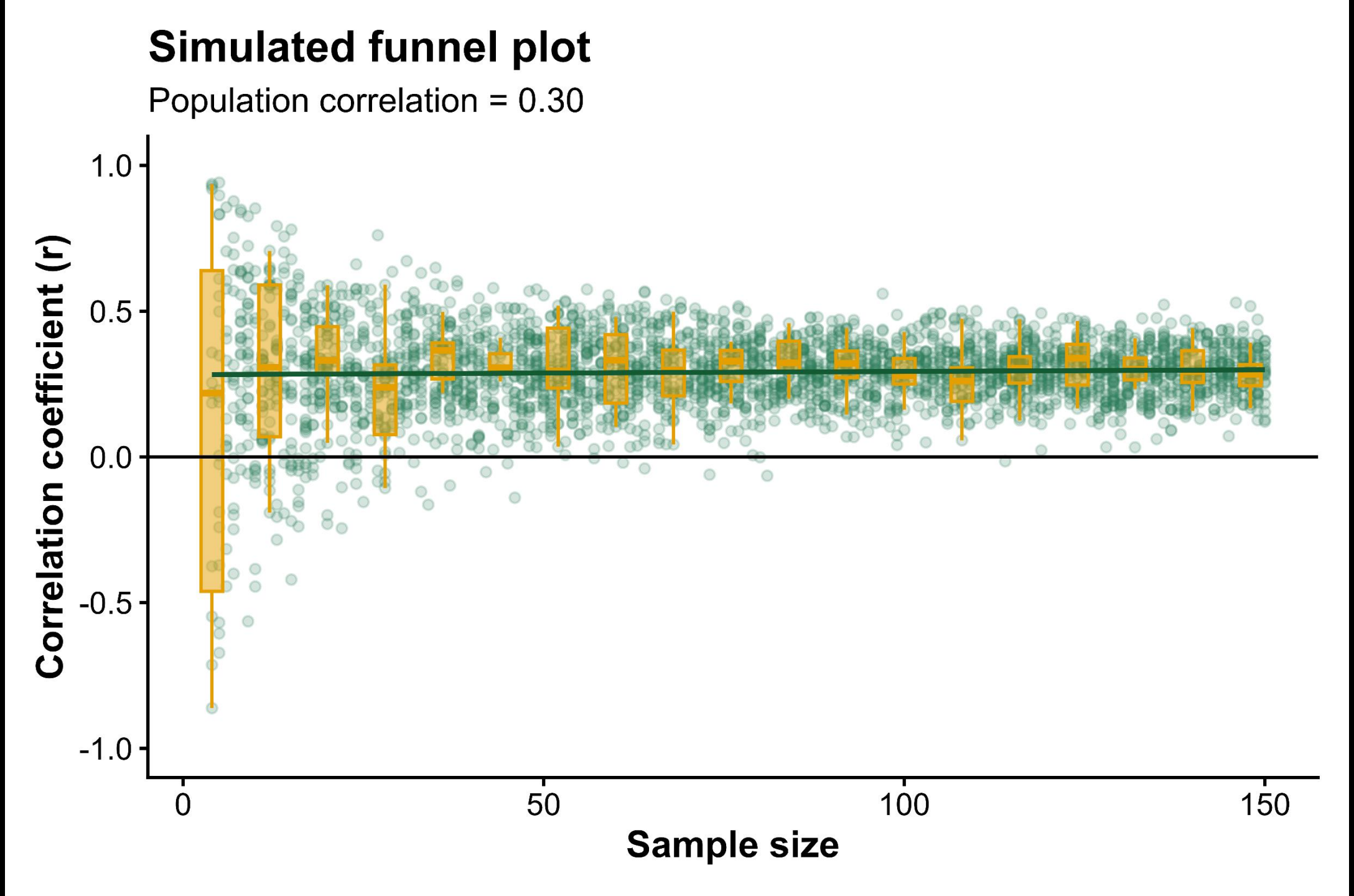
r = 0.08, 95% CI [−0.08, 0.24]

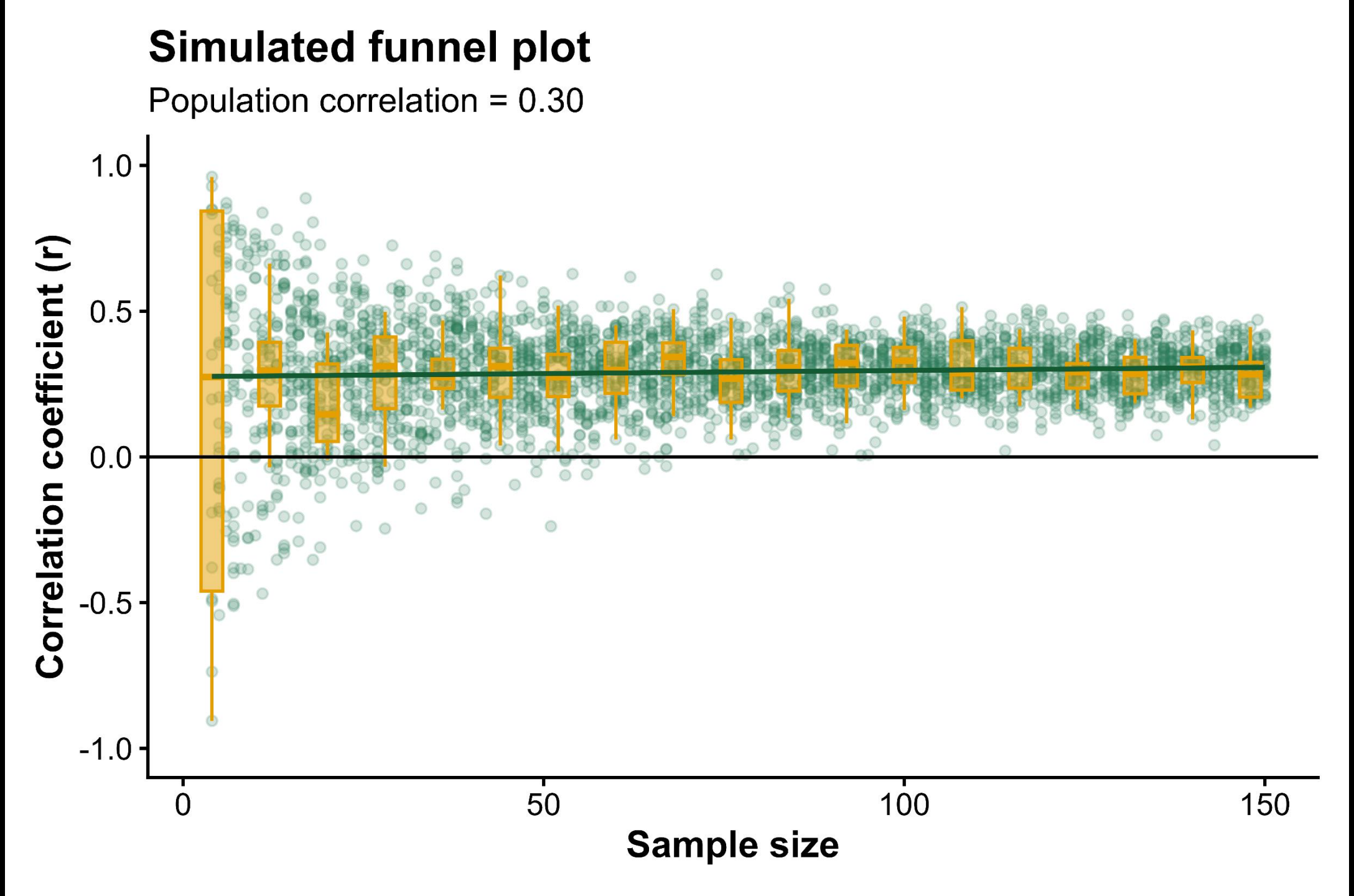


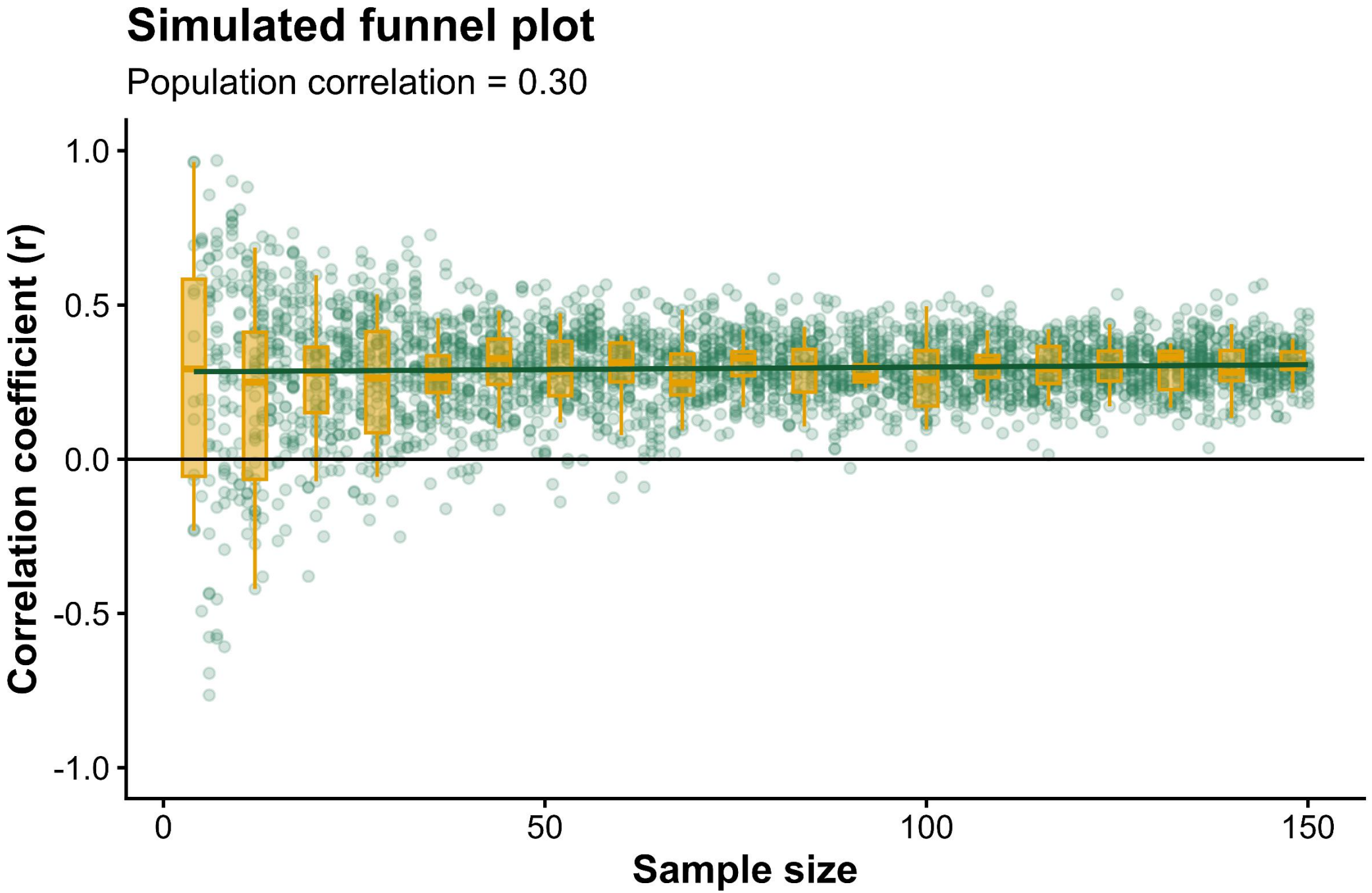

Simulated funnel plot
Population correlation = 0.30
Correlation coefficient (r)
1.0
0.5
0.0
-0.5
-1.0
0
50
100
150
Sample size


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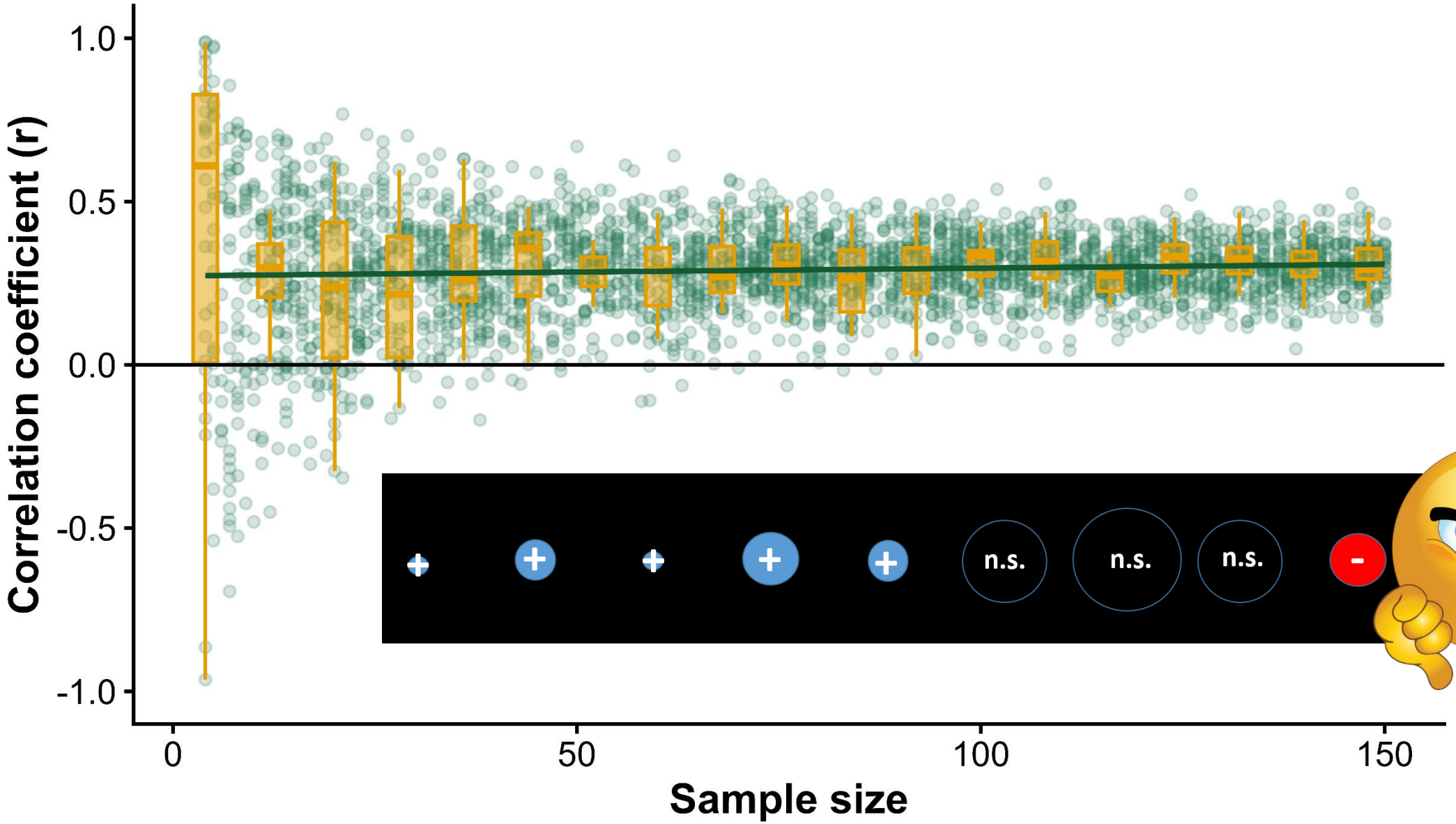

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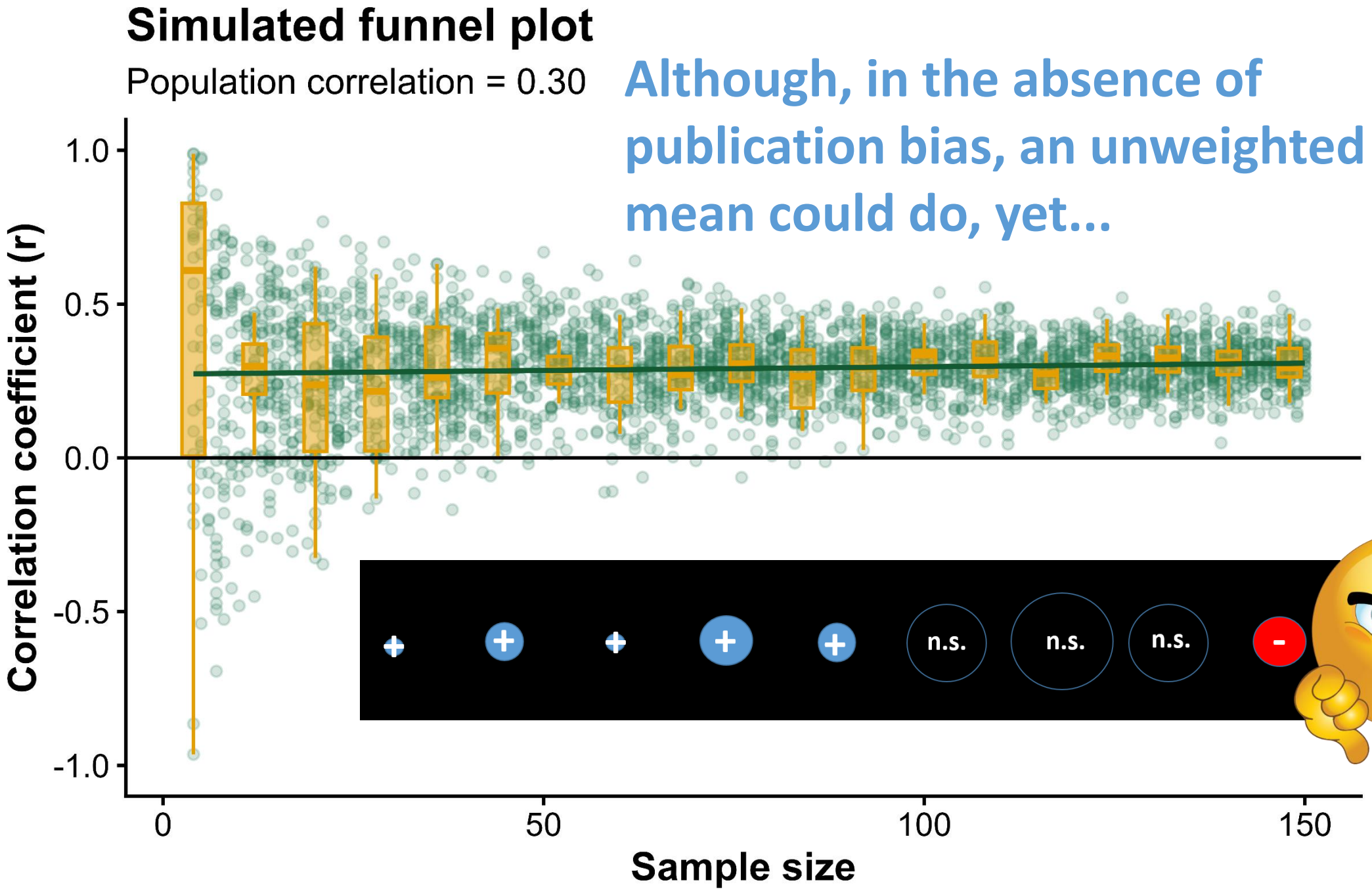

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-0.5
-1.0
0
50
100
150
Sample size

Simulated funnel plot

Population correlation = 0.30





Although, in the absence of publication bias, an unweighted mean could do, yet...

What is this heterogeneity you speak of?

HETEROGENEITY IS THE

Variance in true effects

=

Each dot is one study's effect size, each bar its uncertainty, and the dashed line the mean effect.

What is this heterogeneity you speak of?

HETEROGENEITY IS THE

Variance in true effects

=

SAID DIFFERENTLY

Variability in study outcomes

What is this heterogeneity you speak of?

HETEROGENEITY IS THE

Variance in true effects

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SAID DIFFERENTLY

Variability in study outcomes

=

ALSO THOUGHT OF AS

Inconsistency among effect sizes

What is this heterogeneity you speak of?

HETEROGENEITY IS THE

Variance in true effects

=

SAID DIFFERENTLY

Variability in study outcomes

=

ALSO THOUGHT OF AS

Inconsistency among effect sizes

Low heterogeneity



High heterogeneity



Each dot is one study's effect size, each bar its uncertainty, and the dashed line the mean effect.

What is this heterogeneity you speak of?

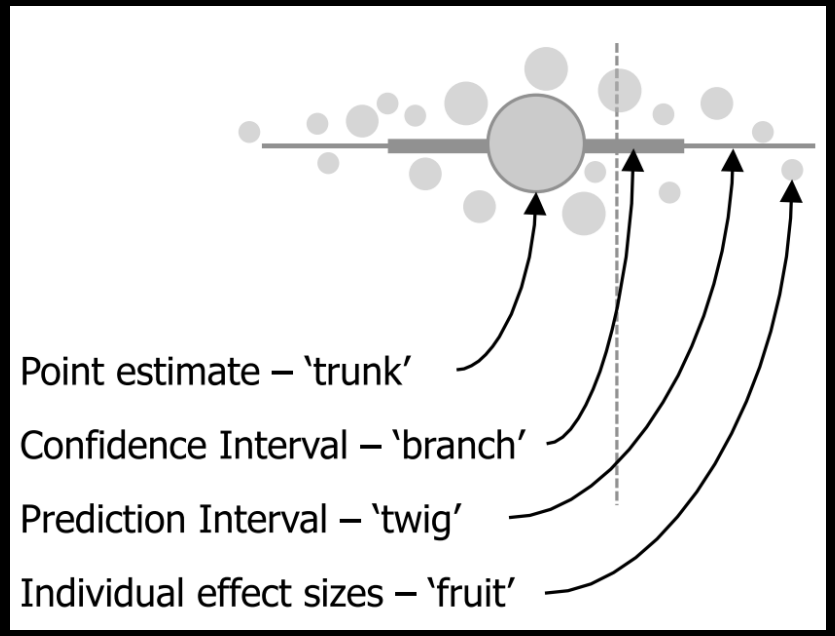
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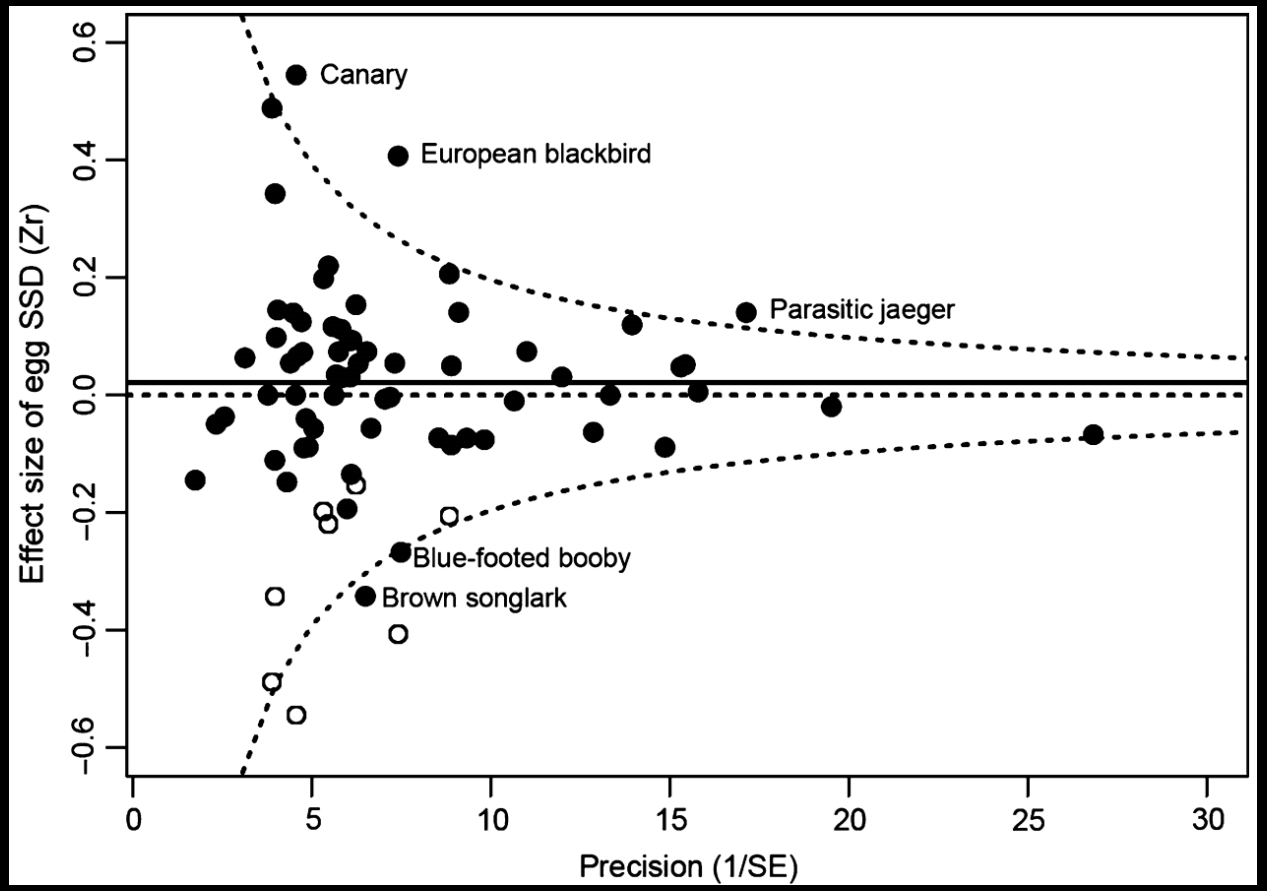
R package 'orchaRd': Nakagawa et al. 2023, MEE

Do females adjust egg size depending on offspring sex?



Rutkowska et al. 2014

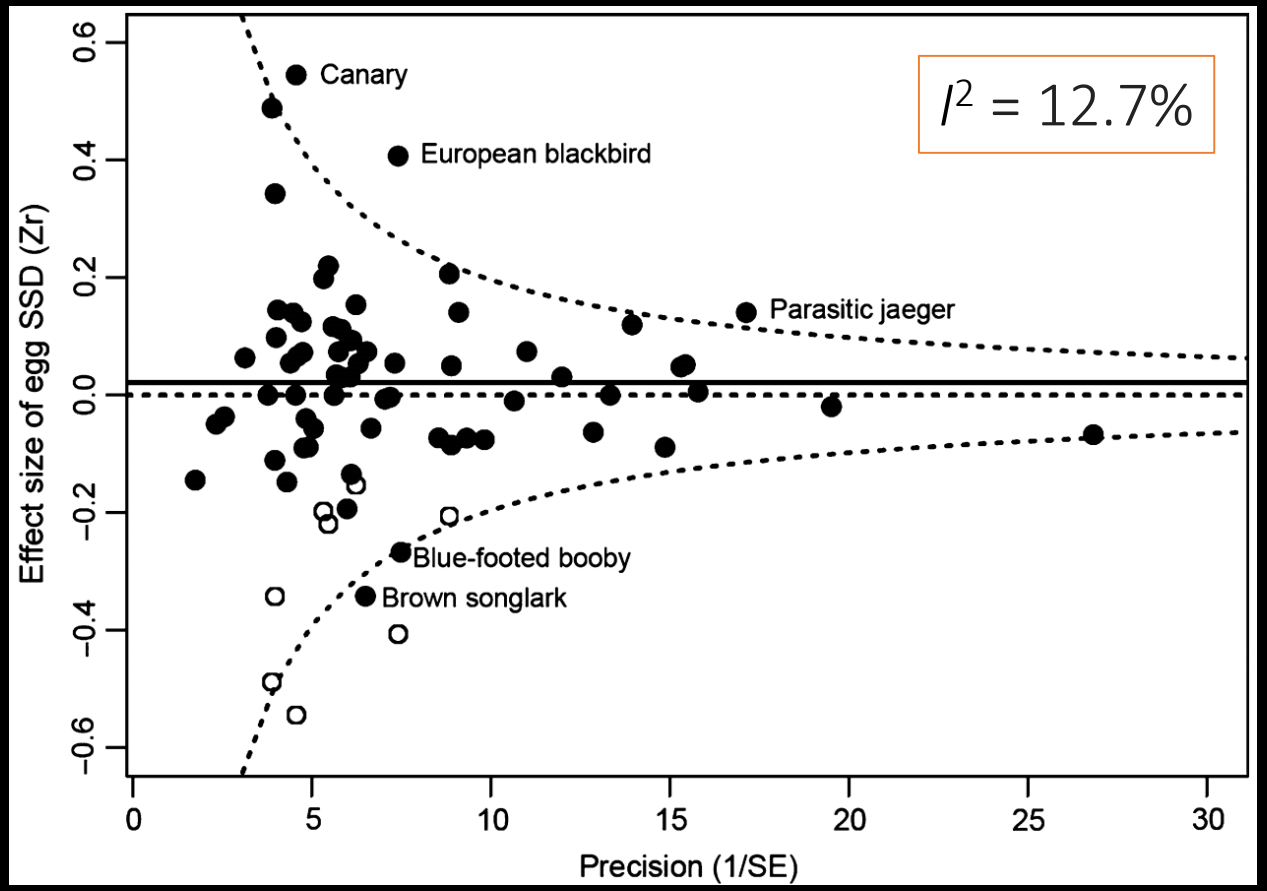
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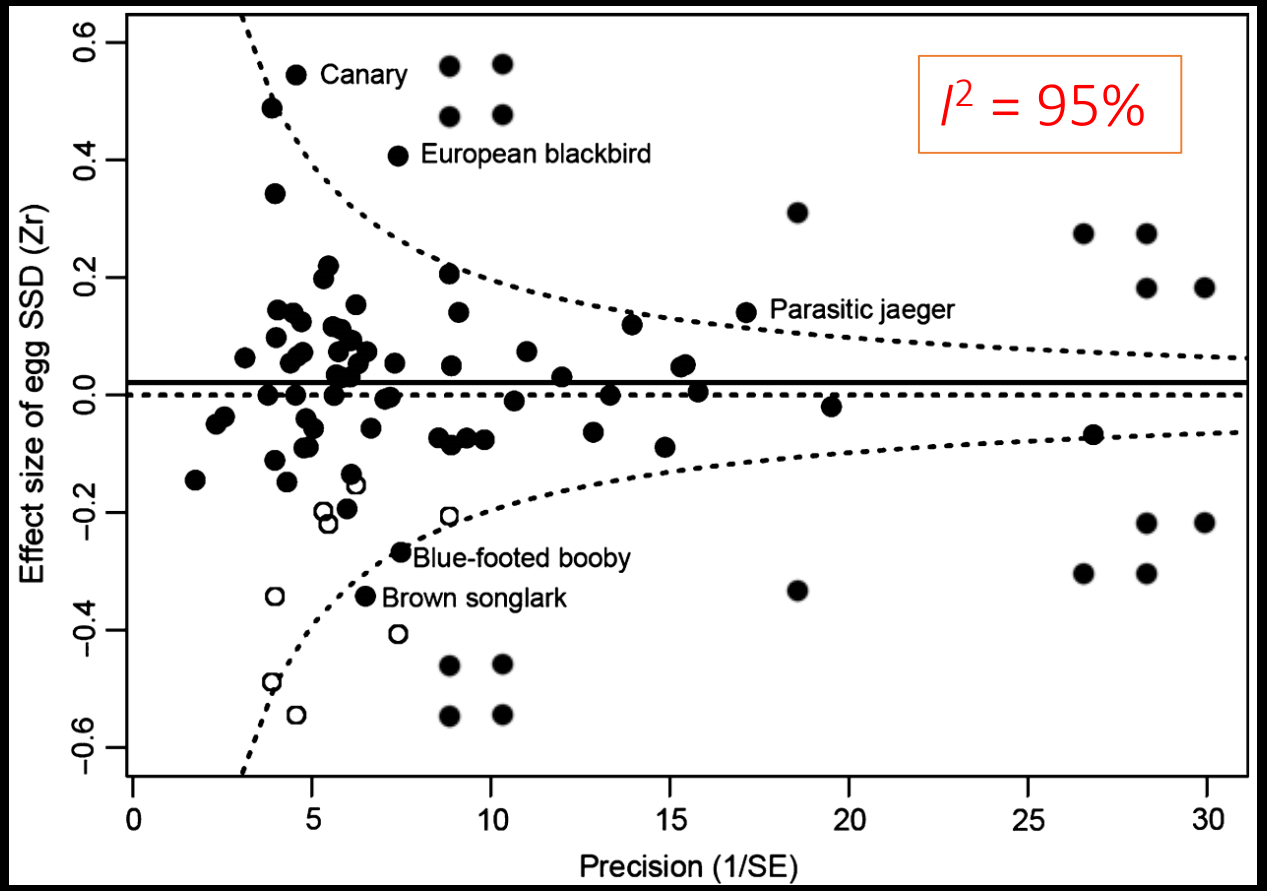
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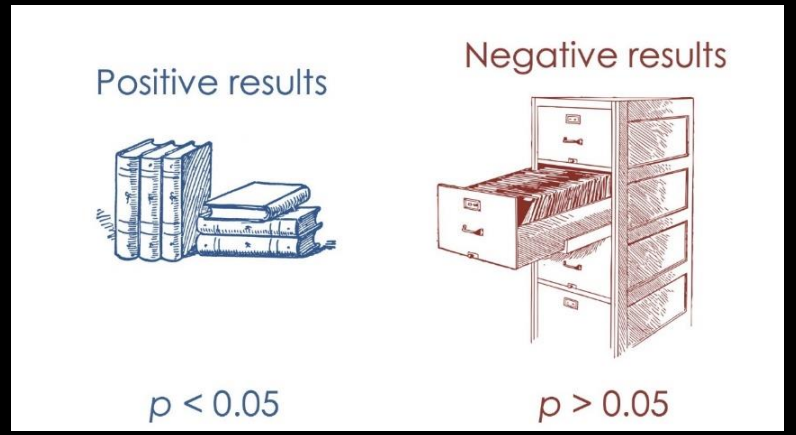
Publication bias

When effect sizes included in a meta-analysis generate different conclusions from an all-inclusive meta-analysis

Reasons:

- Logistical issues (45%)
- Results (e.g., p-value; 25%)
- Review process (20%)
- Problems with study design (5%)
- Goal of the study (4%)



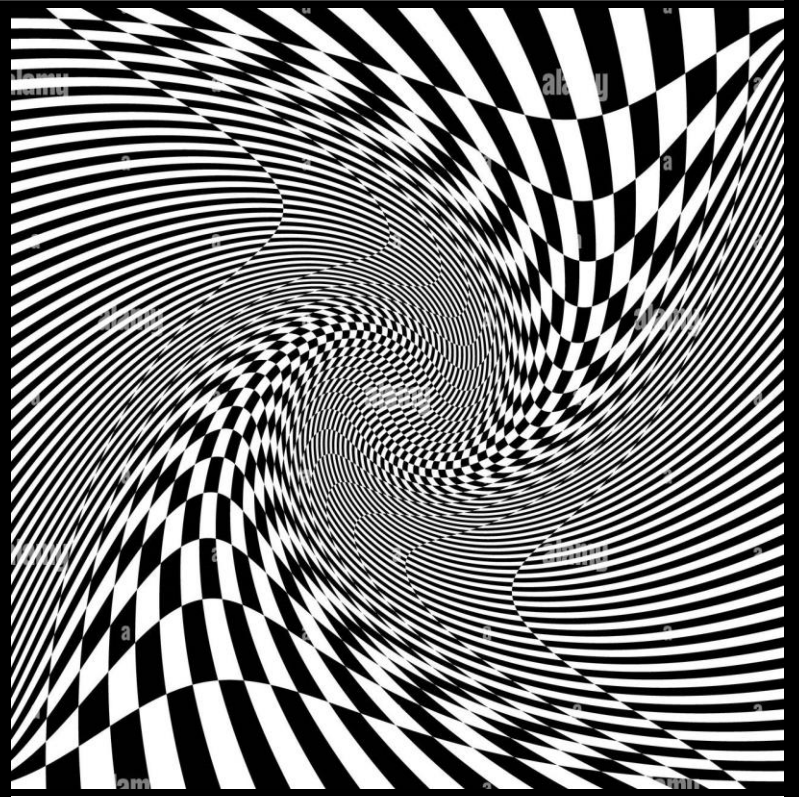


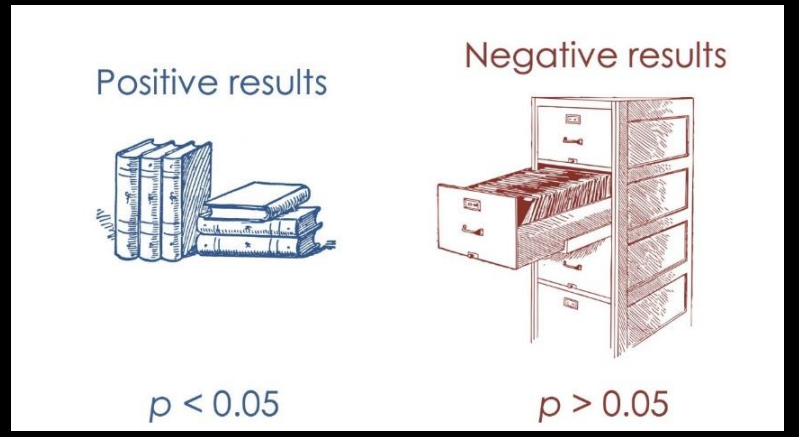
55 respondents; registered studies psychology; Ensinck & Lakens 2024, OSF

Publication bias

When effect sizes included in a meta-analysis generate different conclusions from an all-inclusive meta-analysis

- Small-study effects
- Decline effects

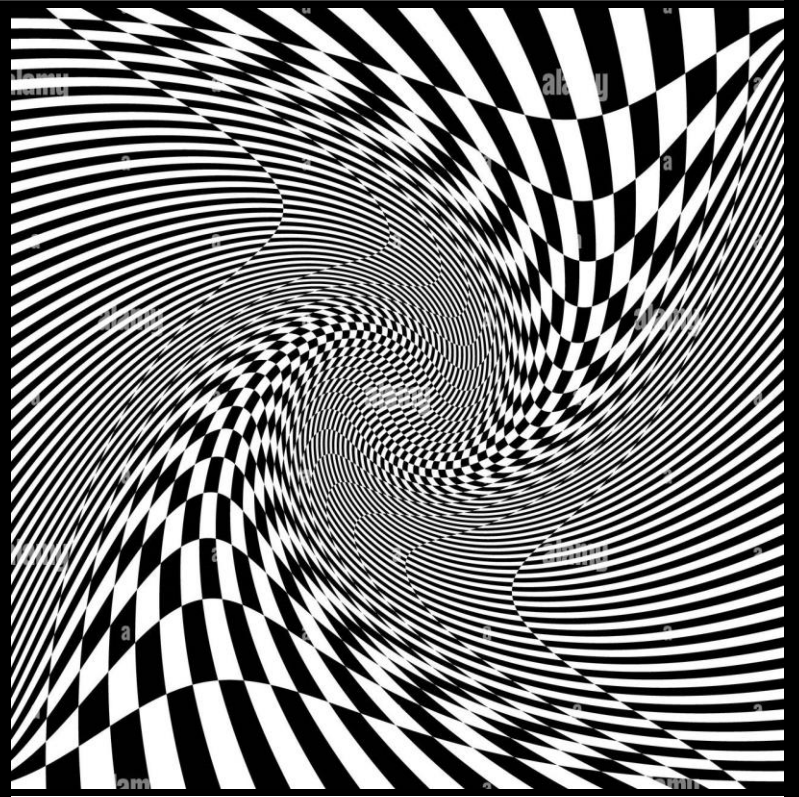


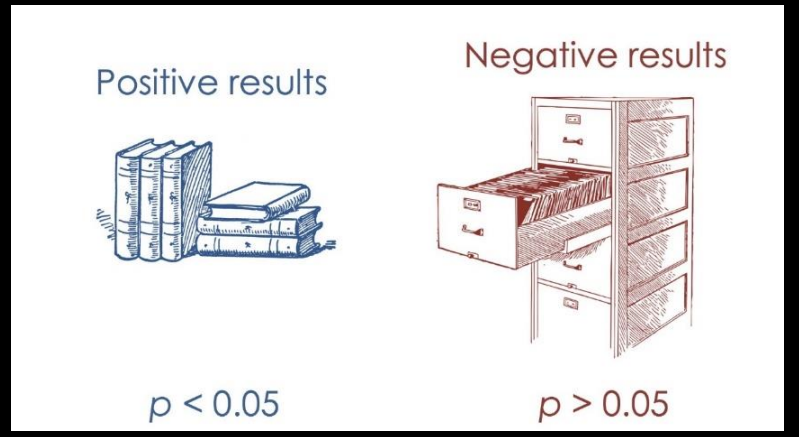


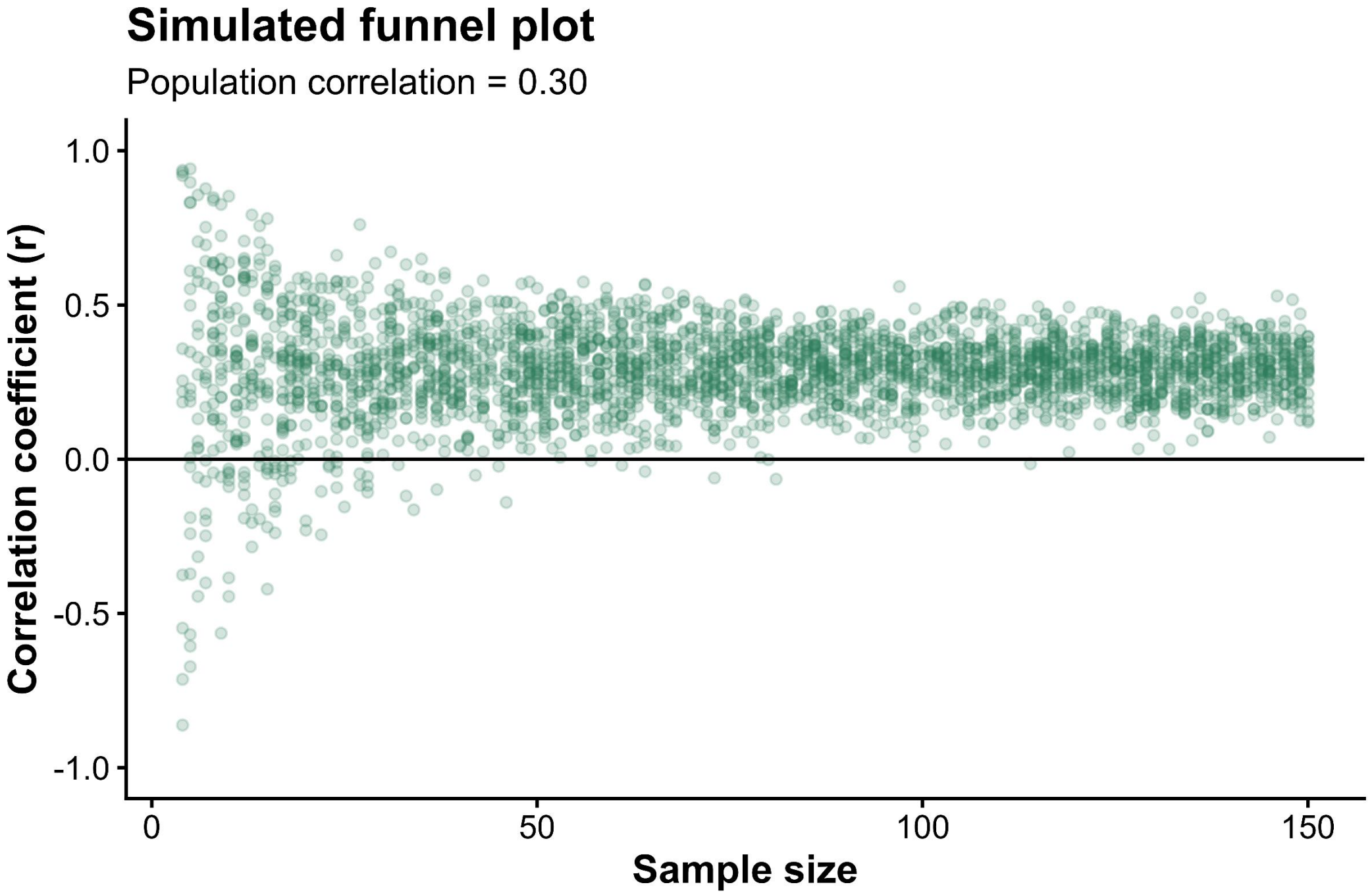
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- Decline effects

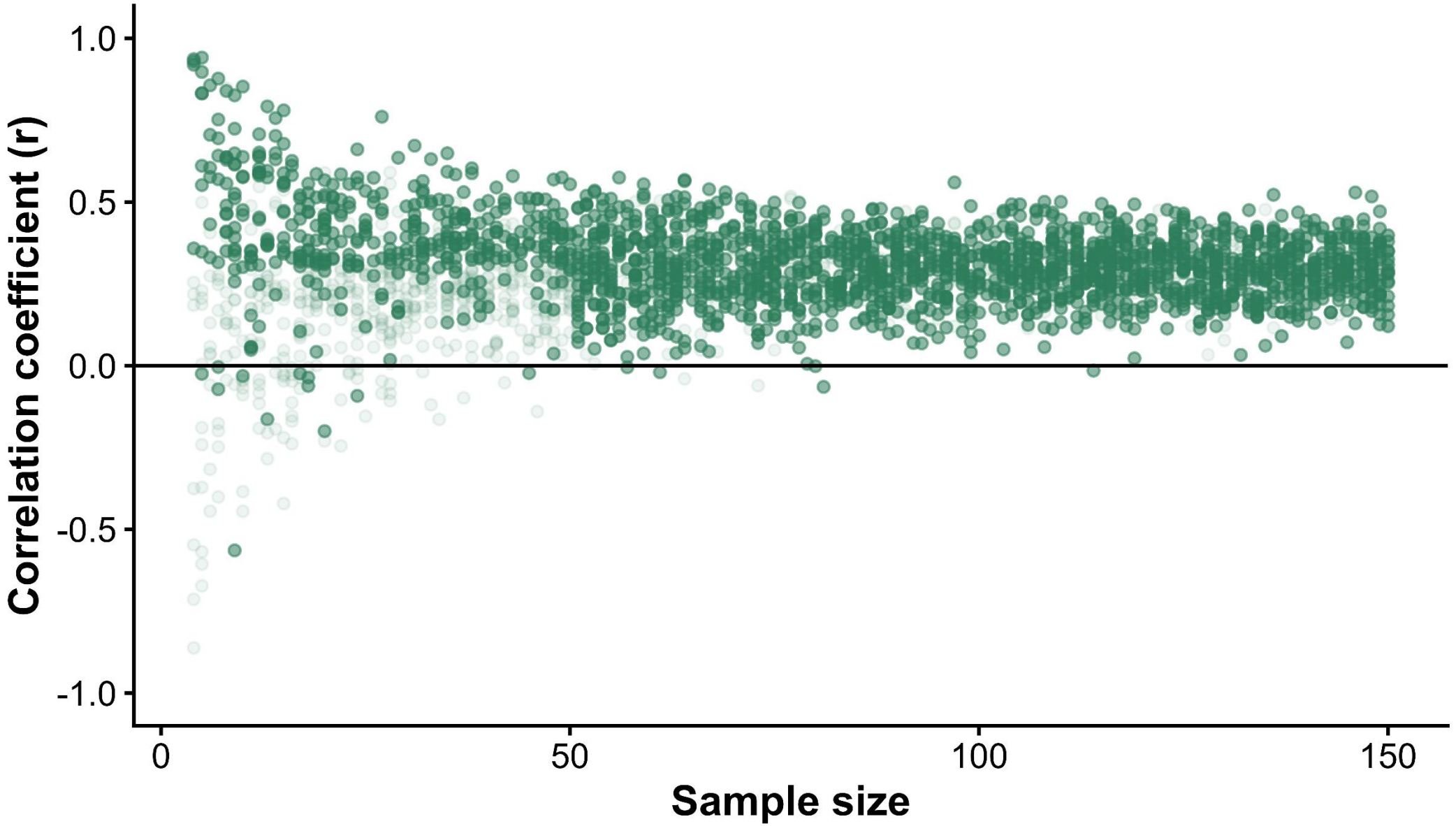


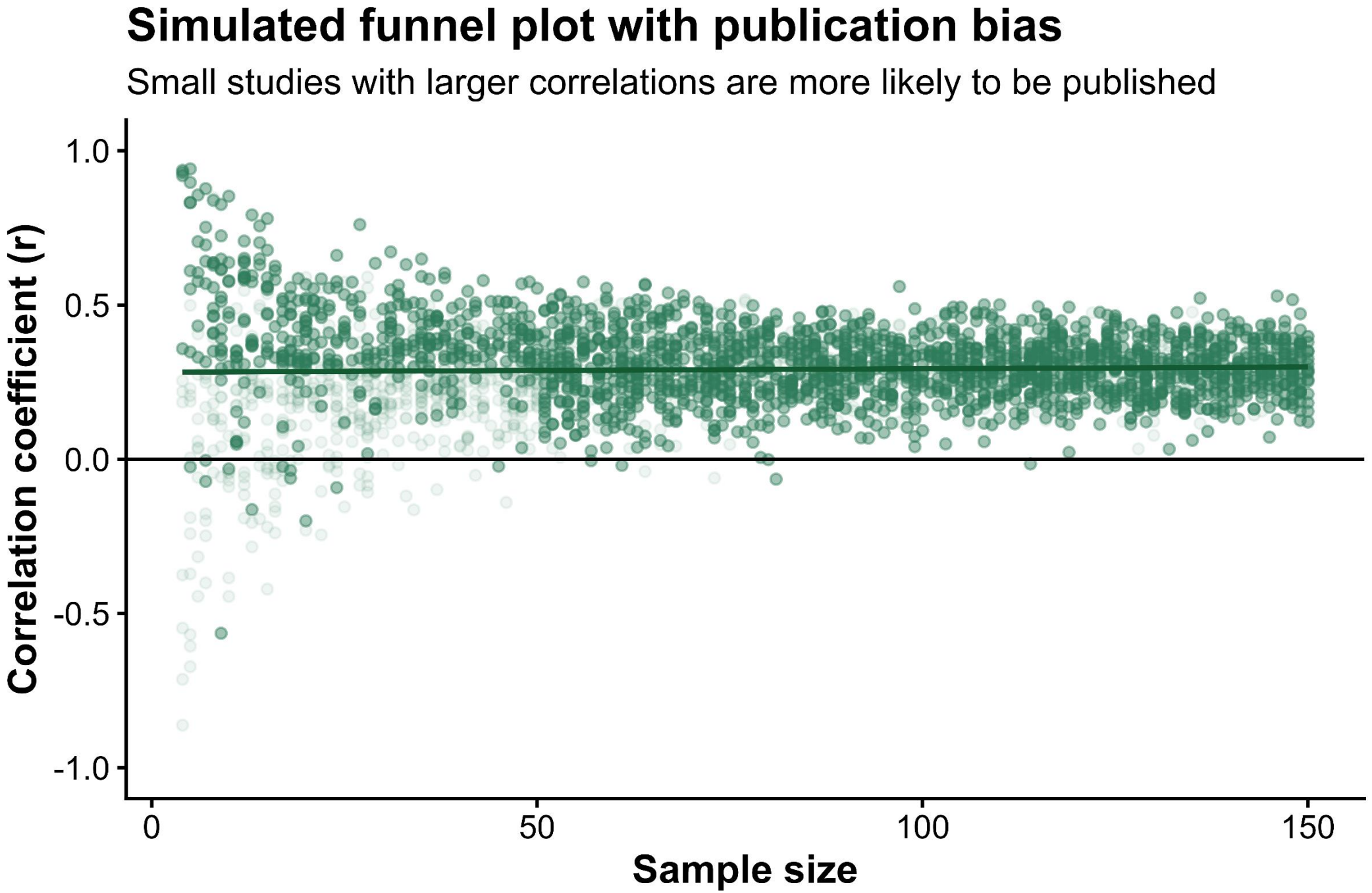



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50
100
150
Sample size

Simulated funnel plot with publication bias

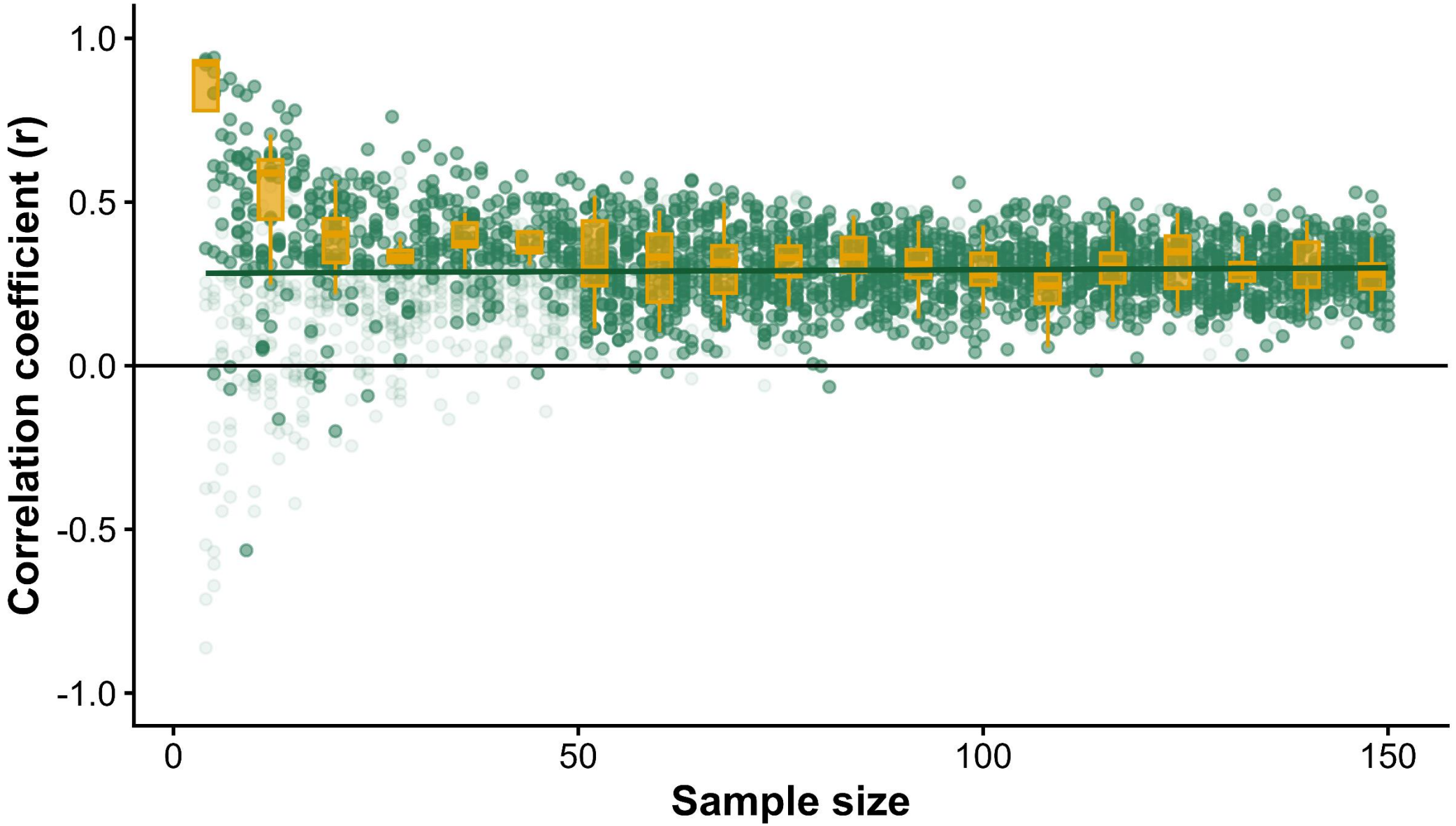
Small studies with larger correlations are more likely to be published

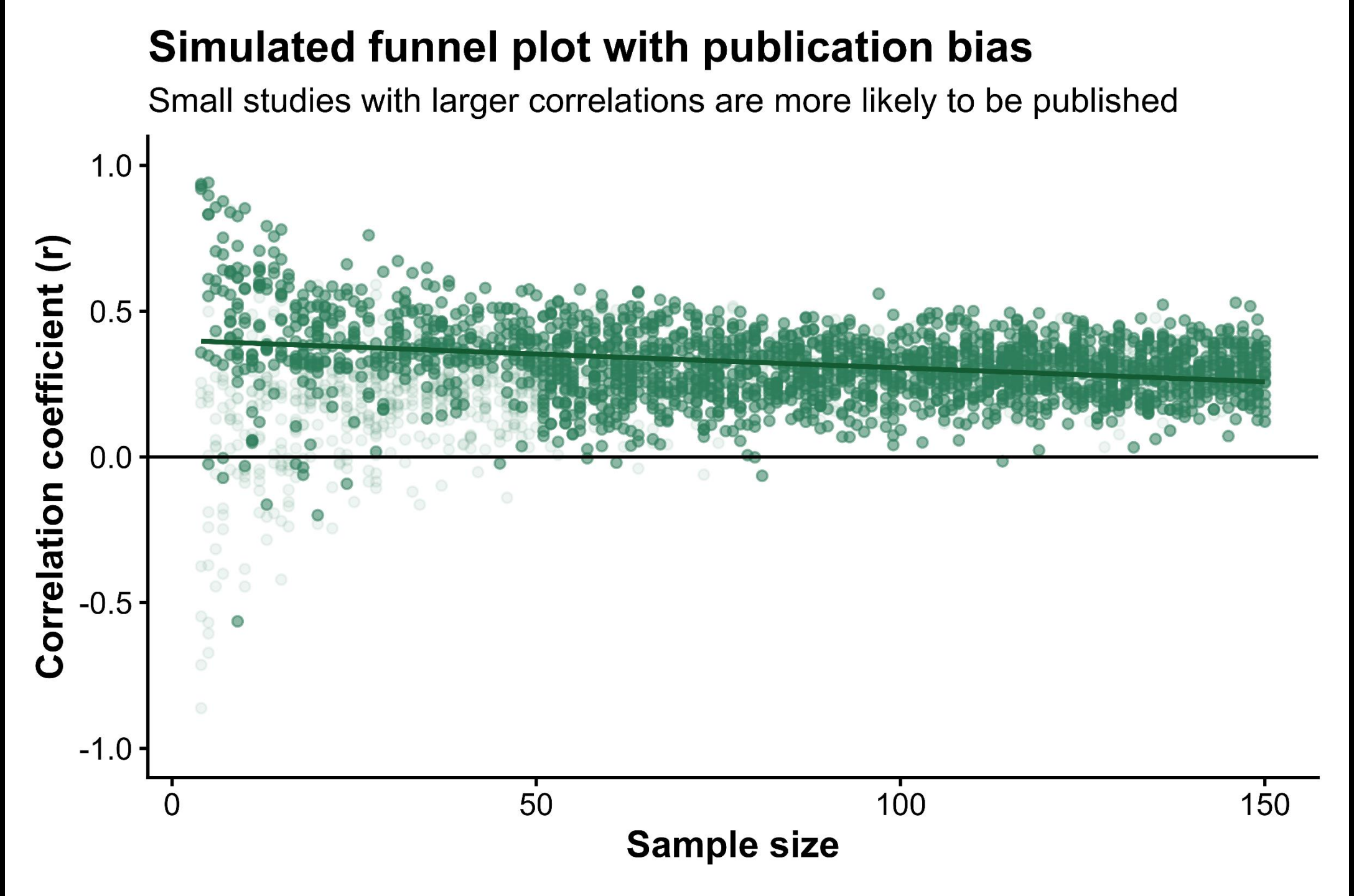



Simulated funnel plot with publication bias
Small studies with larger correlations are more likely to be published
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Simulated funnel plot with publication bias

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Simulated funnel plot with publication bias
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Publication bias: test & adjust for it!

Received: 8 April 2021 | Accepted: 6 September 2021

DOI: 10.1111/2041-210X.13724

Methods in Ecology and Evolution | BRITISH ECOLOGICAL SOCIETY

REVIEW ARTICLE

Methods for testing publication bias in ecological and evolutionary meta-analyses

Shinichi Nakagawa[1] | Malgorzata Lagisz[1] | Michael D. Jennions[2] | Julia Koricheva[3] | Daniel W. A. Noble[2] | Timothy H. Parker[4] | Alfredo Sánchez-Tójar[5] | Yefeng Yang[1] | Rose E. O'Dea[1]

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Review of 102 meta-analyses

